

Web-Based Augmented Reality  
Technologies in E-commerce:  
Evolution Analysis and  
Implementation Through Virtual Shoe  
Try-On

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# 1. Introduction

## 1.1 Background

The increased usage of Augmented Reality (AR) in the web environment has changed how users interact with online environments including e-commerce, fashion, and gaming. The integration of AR into websites enhances user experience by making immersive interactions and personalized encounters. Users can visualize the product in their daily lives, through the enhanced expectations created by AR technology.

Virtual try-on experiences have been around for quite some time, however, only recently have they gained widespread recognition. This is not a new technology— several major retailers, such as Nike and Amazon, have already implemented and actively use AR-enhanced features in the market. These features allow customers to virtually try on products like shoes and eyewear.

The main goals are to reduce uncertainty regarding product fit and appearance, promote user interactivity, and decrease return rates are the main objectives. The idea of virtually trying on products from one's home is becoming more trendy, especially in the post-COVID digital environment, where actual store visits are disintegrating.

This research project is created based on the designs and evaluation of the web-based AR system for virtual shoe try-on. The aim is to investigate how far this technology advances the interactive, fun, and realistic experience in virtual shopping contexts. The system will operate entirely via web browsers; therefore no application installations are going to be needed. By building a prototype, the study examines the evolution of Web AR, the major technical difficulties surrounding its deployment, and the impacts it has on both the user experience and online retail performance.

## 1.2 Aim and Objectives

The aim of this project is to develop and evaluate a web-based virtual shoe try-on platform using augmented reality, with a focus on user experience, technical performance, and accessibility across modern devices and browsers.

The key objectives of the project are:

- To research how AR will change over the course of time in online retail, particularly over the period of 2015 through to 2025.
- To study and investigate the technical requirements and challenges in implementing AR-based virtual try-on website.
- To develop a functional prototype that supports 3D shoe visualization and real-time AR try-on capability using third-party tools like Wearfit.
- To conduct user testing and performance evaluation in terms of technical metrics (loading time, frame rates, memory usage) and user feedback (satisfaction and engagement).
- To present how easy or complex it is to implement WebAR, particularly for small or individual developers using modern web frameworks like Next.js, and ready-made tools such as Wearfit.

## 1.3 Research Questions

This study will address the following research questions:

- *How has web-based AR technology evolved in e-commerce applications from 2015 to 2025?*
- *What are the current technical challenges and solutions in implementing virtual try-on features for footwear?*
- *How does AR implementation influence user engagement, satisfaction, and purchase confidence in online shopping?*

- *How do various devices and browsers affect the performance, compatibility, and user experience of WebAR try-on features?*

## 2. Literature Review

The research in this literature review is focused on the evolution, technical progress, and user impact of web-based augmented reality technologies in ecommerce, with a specific focus on virtual footwear try-on website. This review uses a thematic approach, with the structure of the review matching the research questions that the study sets out to address.

The academic sources for this study were collected from trusted databases such as Google Scholar, Edinburgh Napier LibrarySearch, SpringerLink, and ScienceDirect. Industry reports and technical articles, such as BrowserStack reports on cross-browser issues and CatchMetrics studies on mobile performance, were also used to support the research. The main search keywords included “WebAR in ecommerce,” “virtual try-on footwear,” “WebXR,” and “3D shoe visualization,”

The literature review focus on studies published within the time frame from 2015 to 2025. However, it is also included a few old and significantly meaningful research because of its relevance and usefulness for providing necessary background for understanding augmented reality and virtual try-on technology. Selected materials included peer-reviewed journal articles, case studies, and industry examples including Nike Fit, and Adidas. This bridged the gap between empirical and theoretical knowledge and practice.

The literature review can be outlined into following structure -

- 2.1 What is AR and WebXR? –
- 2.2 Evolution of AR Technology in E-commerce –
- 2.3 AR Tracking Methods and Technical Foundations
- 2.4 Impact of AR on Online Shopping Behaviour
- 2.6 Implementation Strategies and Retail Case Studies
- 2.7 Challenges in Implementing AR in E-Commerce
- 2.8 Future Directions and AI-Enhanced Experiences
- 2.9 Addressing the Research Questions

### 2.1 What is AR and WebXR?

Augmented Reality (AR) is a technology that adds digital content to the real world, letting users see virtual objects in their physical surroundings (Nikolaidis, 2022). AR is used in many areas like education, healthcare, entertainment, and retail, and its growth has been supported

by web technologies such as WebXR, which allows people to use AR directly through their web browsers.

Web Augmented Reality (WebAR) is a technology that lets users see digital content in the real world using just a web browser, meaning no other standalone apps are needed. This makes shopping more interactive and helps connect online and in-store experiences. According to [Jeganathan \(2021\)](#), WebAR is an important step forward in AR, as it allows businesses to offer immersive experiences easily, without requiring customers to download anything.

WebAR enhances user engagement by providing an interactive interface that allows consumers to visualize products in their own environment. This kind of aspect is essential for e-commerce because usually interactive touchpoints are often limited in them. These integrations support the findings of Gatter et al., which show that AR technology meets what customers want to touch, feel, and experience products before making a purchase in a virtual setting, increasing the appeal of online shopping for customers who prefer a tangible product experience ([Gatter et al., 2021](#)). Furthermore, as Riar et al. point out, the introduction of AR to the retail industry has resulted in significant behavioral and psychological changes that have a beneficial impact on consumer decision-making processes by raising engagement and satisfaction ([Riar et al., 2022](#)).

## **2.2 Evolution of AR Technology in E-commerce**

In early development of Augmented Reality (AR), due to a lack of technological advancement and a confused user experience, early e-commerce implementations of AR faced major adoption difficulties. For example, to use AR technology in earlier days users required desktop systems and standalone mobile applications and they had to download additional software to run on them. This created the shopping process more difficult. [Van Krevelen and Poelman \(2010\)](#) identified that these systems demanded specific hardware configurations, including webcams and high-performance devices, therefore only people with devices with advanced technology could use them.

According to [Kim and Forsythe \(2008\)](#), this app- or desktop-dependent approach created barriers to technology adoption, as users frequently stopped using AR because of the additional steps required to access it. Additionally, some studies state that technology anxiety was one of the reasons why users may experience discouraged feeling from using virtual try-on technologies. According to [Poushneh and Vasquez-Parraga \(2017\)](#), these obstacles compiled simple processes, interfering the smooth purchasing experience. Hilken et al. (2017) also noted that these extra processes considerably reduced user interest, even though AR has the potential to improve product visualization and boost buy confidence.

Another issue discussed was compatibility problems in AR adoption with users finding it harder to get comfortable with this technology. Guo and Zhang (2024) observe that many of

the early AR applications failed to properly communicate with the online shopping platforms therefore the entire shopping experience is disconnected. Yim et al. (2017) also states these early systems needing high-end devices causes problems to AR users with low-performance computers or phones which also align with Van Krevelen and Poelman (2010). Consequently, users faced problems such as needing to download additional apps, going through extra setup and adjustments, and having high-performance devices. While the technology of AR has a lot of potential to provide better user experience, these barriers blocked it from becoming widely used in online shopping.

### **2.3 Rise of Mobile AR and Industry Applications (2015-2019)**

Between 2015 and 2019, advances in mobile technology, including improved cameras and processing power, helped the growth of mobile AR applications. [Dacko \(2017\)](#) reported that the number of AR shopping apps increased from around 30 in 2015 to over 120 by late 2017. This growth was driven by improvements in image recognition, rendering engines, and motion tracking ([Ong et al., 2019](#)).

Despite these advancements, mobile AR still faced usability challenges. Smaller screen sizes, limited device performance, and the need for separate app downloads continued to hinder widespread adoption ([Liu et al., 2024](#)). Nonetheless, the footwear industry became one of the early adopters of AR in online shopping, with brands like Converse and Nike among the first to develop AR-based virtual try-on applications for footwear, as seen in Converse's The Sampler app (2010) and Nike's Nike Fit feature (2019) ([Communication Arts, 2010](#); [Engadget, 2019](#)). While these applications helped reduce purchase uncertainty, the need for app downloads still presented and created friction in the user experience ([Ntumba et al., 2023](#)).

### **2.4 Transition to WebAR and Technological Breakthroughs (2018-2020)**

The period between 2018 and 2020 marked a key stage in AR development with the introduction of WebAR technology, allowing users to access AR features directly through web browsers without downloading dedicated applications. This transition significantly reduced barriers to access and expanded the potential user base for AR in e-commerce.

Some of the studies highlighted that WebAR improved accessibility compared to desktop and mobile AR applications, although challenges related to screen size and device performance persisted ([Ong et al., 2019](#)). The introduction of WebGL in 2011 laid the foundation for browser-based 3D rendering, but true AR integration was enabled by the WebXR Device API, standardized from 2018 onwards ([Muff & Fill, 2024](#)). This technology unified fragmented AR systems and allowed users to move smoothly from browsing to interactive AR experiences within their web browsers.

In addition, WebAR's impact on the retail sector was further supported by the release of ARKit by Apple and ARCore by Google in 2017, providing developers with tools for creating AR features that worked across devices ([Diana et al., 2023](#)). Dacko (2017) noted that advances in browser technology led to the rapid growth of AR shopping applications, with WebAR removing previous adoption barriers associated with app downloads.

In conclusion, the evolution of AR technology in e-commerce shows a clear shift from complex, app-based systems to more accessible and user-friendly WebAR solutions. Early AR adoption faced significant barriers due to hardware requirements, app downloads, and poor integration with e-commerce platforms. The rise of mobile AR improved accessibility but still presented usability challenges. The introduction of WebAR between 2018 and 2020 marked a turning point, removing major access barriers and enabling seamless AR experiences directly through web browsers. This advancement has allowed businesses, particularly in the footwear industry, to enhance customer engagement and improve the overall online shopping experience.

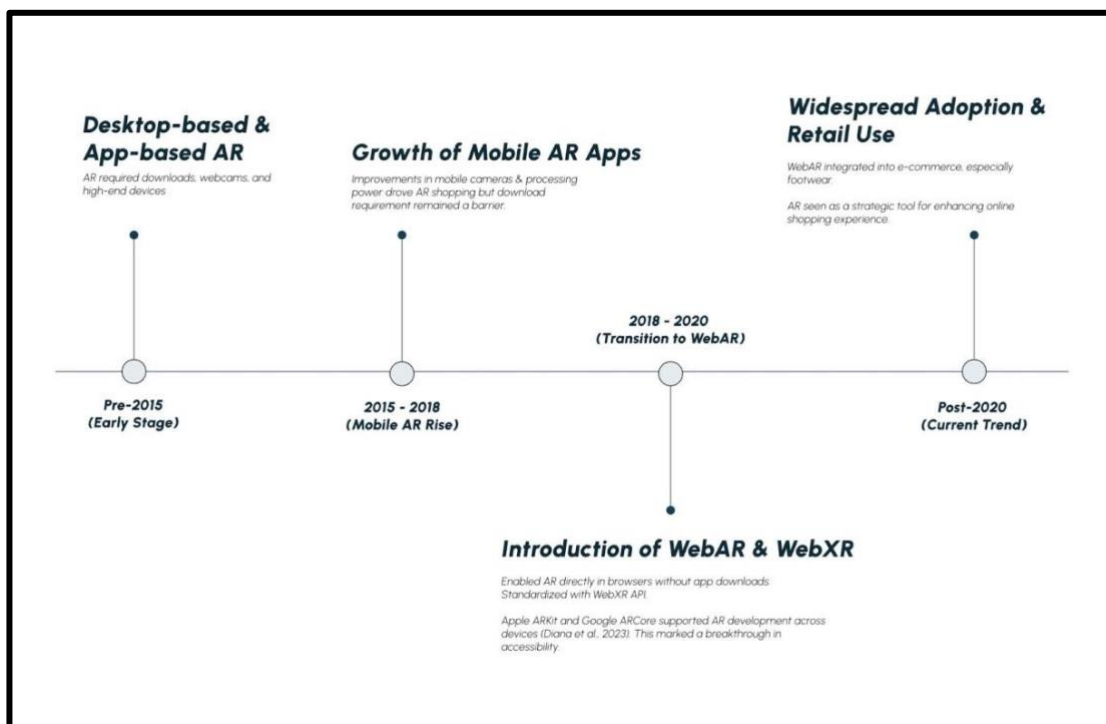


Table 2.1: Evolution of AR Technology

**Table 2.2: Reality of AR Challenges Today**

| Problem                          | Is it still a challenge?    | Why/Evidence                                                                                                                                                                                                                                           | Supporting Studies                                        |
|----------------------------------|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| Device Performance               | <i>Partially exists</i>     | <p>Low-end and mid-range phones could struggle with resource heavy 3d Models.</p> <p>High-performance phones can handle AR well.</p>                                                                                                                   | <p>CatchMetrics (2025);</p> <p>Liu et al. (2024)</p>      |
| Cross-Browser Compatibility      | <i>Still exists</i>         | <p>WebXR is not fully supported across all browsers, noticeably Safari.</p> <p>Chrome and Android are better at displaying AR content.</p>                                                                                                             | <p>BrowserStack (2025);</p> <p>Muff &amp; Fill (2024)</p> |
| Foot Tracking Accuracy           | <i>Still a challenge</i>    | <p>AR shoe try-on is good at visualization and not at precision fitting.</p> <p>Accurate sizing &amp; fit tracking is still experimental, especially in WebAR. Big brands (Nike, Adidas) use AI + AR to improve this, small brands could struggle.</p> | <p>MDPI (2021);</p> <p>Chen &amp; Lee (2022)</p>          |
| Technical Development Complexity | <i>Exists but improving</i> | <p>Platforms such as 8thWall, Zappar, and Shopify AR tools make AR integration easier.</p> <p>Fully customized WebAR still requires skilled developers for performance optimization.</p>                                                               | <p>Plutomen (2020);</p> <p>Muff &amp; Fill (2024)</p>     |

## **2.5 Impact of AR on Online Shopping Behaviour**

This section of the literature review explores how WebAR implementations impact key aspects of consumer behavior within footwear e-commerce. In particular, the review will examine how AR enhances interactivity, increases purchase intention, reduces purchase uncertainty, fosters emotional engagement, and supports cognitive decision making processes. Furthermore, this section will highlight how the technological characteristics of AR — such as interactivity, vividness, and sensory feedback — contribute to improved user experiences, customer satisfaction, and reduced product returns in the context of online footwear retail.

### **2.5.1 Enhanced Purchase Intention**

Research indicates that augmented reality technology has a beneficial impact on consumers' purchase intentions. [Adam and Pecorelli \(2018\)](#) investigated, for instance, how user decisions might be greatly influenced by seller recommendations and customer reviews included into augmented reality applications. Customers are more likely to feel confidence in their decisions and make a purchase when they interact with augmented reality (AR) in products and get positive feedback. This is similar to the general finding in the literature that higher buy-in and decreased uncertainty are facilitated by enhanced social proof offered through AR ([Riar et al., 2021](#)).

### **2.5.2 Reduced Purchase Uncertainty**

[Kowalczyk et al. \(2021\)](#) examined the role AR and state that it is beneficial in reducing uncertainty during purchasing behaviours. One of its advantages is letting consumers view the product in the situation and environment where it is expected to be used—for example, furniture in their own homes or virtually trying on clothes. Undoubtedly, by portraying a more accurate picture of how products will work and look in real life, AR helps reduce the likelihood that they would be unhappy with the product after making a purchase. This kind of confidence is critical for motivating a consumer to actually complete the purchase in the case of expensive items.

### **2.5.3 Emotional Engagement and Enjoyment**

Shopping is made more enjoyable and emotionally engaging by AR in addition to being made simpler. According to [Hilken et al. \(2017\)](#), the immersive aspects of AR can increase people's excitement and enjoyment of the shopping experience. Customers are more likely to return and make more purchases when they have a positive shopping experience.

### 2.5.5 Technological Characteristics of AR

The technological features of AR—such as vividness, interactivity, and sensory feedback, improve both hedonistic and utilitarian buying experiences. Several studies examined by [McLean and Wilson \(2019\)](#) confirm such AR features offer positive value on consumer behavior by making shopping entertaining and educational. For example, interacting with products via augmented reality enhances user satisfaction and likelihood of recommendation.

In summary, the section highlights that AR technology plays a significant role in shaping online shopping behaviour, particularly in the footwear industry where product visualization and fit accuracy are critical. By providing interactivity in digital products, AR improves user engagement and encourages consumer confidence, hence increasing purchase intention. By allowing people to try virtual products, AR effectively reduces purchase uncertainty. All these effects enhance customer satisfaction and may also reduce product return rate. This is why AR is seen as a major enhancement to the online shopping experience.

### 2.5.6 Technical Implementation Considerations

However, the initial loading speed represented the most critical performance factor for footwear WebAR adoption, with consumers abandoning experiences that took longer than 3 seconds to initialize (BrowserStack, 2024).

In addition to a technical perspective, various studies have stressed the importance of leveraging specialized AR technologies in enhancing user experiences in footwear e-commerce. [Ong et al. \(2019\)](#) explored the effectiveness of AR shopping applications and suggested that AR effects such as foot tracking deliver better results when contextualized to the situation of use. They found that customized AR functions do engage the customer better and have a greater impact on purchase intention than general-purpose AR tools. [Liu et al. \(2024\)](#) similarly looked at how AR technology assists in changing consumer shopping behavior with respect to footwear. It was found that dedicated technologies like specialized foot-tracking libraries provide enhanced user interaction and create a more seamless and confident shopping experience. These studies show that the targeted technological development is required to exceed usability concerns and promote accurate and engaging AR experiences in online footwear retail.

## 2.6 Implementation Approaches for AR in Retail

Many footwear brands have used AR technologies effectively to improve the customer journey. For example, these companies launched AR apps to allow consumers to virtually try on shoes on their mobile phones, making an interactive shopping experience that enhances customer engagement (McLean & Wilson, 2019).

In other places, retail chains have installed AR displays in brick-and-mortar stores to facilitate immersive product visualization aided by real-time insights (Scholz & Duffy, 2018).

There are several strategies of implementing augmented reality (AR) technologies when retailers adopt implementation to provide retailers with enhanced shopping experience in the context of virtual try-ons. Collaboration with AR technology vendors such as Wanna, Vyking and Shopify allows retailers to use readily available frameworks and toolkits to deliver seamless virtual fitting experiences.

Another interesting example is Walmart, along with other big retailers, has been actively acquiring augmented reality start-ups such as Zeekit to develop AR solutions that promise greater customer engagement and satisfaction (Perannagari & Chakrabarti, 2019; Walmart Inc., 2022a). Retailers also have the choice of implementing semi-custom setups using AR frameworks such as ARCore and ARKit, which ease the development of applications tailored for specific customer needs and preferences (Perannagari & Chakrabarti, 2019).

In addition, white-label solutions allow timely implementation of AR features such as virtual try-ons, thereby keeping expenditures low, and maintaining brand identity. The combination of various approaches to implementation in collaboration with partners, acquisitions for specific purposes, semi-custom development, and whitelabeled solutions all enable retailers to enhance their offerings in the ever-competitive e-commerce arena and thereby offer more interactive and fun shopping experience for consumers (Xue et al., 2024).

## **2.7 Case Studies of Successful AR Implementation**

Nike's WebAR application for shoes shows the effectiveness of such integrated technology. The Nike Fit uses augmented reality to scan a customer's foot using a smartphone camera and collects 13 data points to create the model of the customer's feet. From that point on, the model is used to recommend shoe size as well as style, improving online shoe fitting accuracy. This new technology, according to Nike, has reduced online returns by 28%, which equated to significant cost savings and better customer satisfaction ([JeffreyBoopathy, 2021](#); [BrandXR, 2023](#)).

Similarly, the company Adidas allowed its customers to immerse themselves in the mall of the future through 3D models and augmented reality. With 3D views in product pages, customers can view the shoes from all angles and zoom into details and places where shoes where they want to put. This way of interacting with a product seems similar to an in-store shopping experience. Thus increasing a customer's confidence in buying products online ([BrandXR, 2023](#)).

On the other hand, regarding big commercial applications such as Nike and Adidas, many reports have been widely presented on in industry journals and marketing case studies; however peer-reviewed academic research has yet to take up the systematic study of their effectiveness, an in-depth systematic study of its effectiveness, user engagement metrics, or long-term consumer behavior impacts has yet to be undertaken in academic research.

Furthermore, scholarly examination would support these claims and provide a more rigorous understanding of WebAR's role in improving online footwear retailers.

### **2.7.1 IKEA's AR Innovation**

Beyond footwear, the implementation of AR extends in other domains, which is evidenced by IKEA's first solution toward furniture visualization. Their special AR tool, termed 'IKEA Place,' enables consumers to visualize furniture in their real living spaces with smartphones to solve the very basic doubts of shopping for furniture online ([Gatter et al., 2021](#)).

The implementation of IKEA's AR focus is to allow consumers to manipulate the size and position of virtual furniture models to see how products will fit within their homes. This is especially beneficial for consumers who want to touch and experience greater hedonic effects from AR applications as it can result in increased purchase intentions and positive reviews toward products ([Gatter et al., 2021](#)).

IKEA, with the integration of augmented reality technology, overcame one main disadvantage of traditional online shopping: the inability to touch and feel real objects. By doing this, they were able to transform the fundamental aspects of the customer experience. This example serves to further emphasize how AR, when appropriately integrated, can improve consumer satisfaction in an online environment while being able to resolve a specific pain point of that particular category.

Similarly, Adidas improved user engagement by making their WebAR features quick and easy to access, while IKEA used AR to help customers see how furniture would look in their own space, solving common online shopping problems. These case studies show that successful WebAR uses more than just AR technology — they focus on creating smooth user experiences, providing helpful product visualizations, increasing interactivity, and making up for the lack of physical shopping. The positive results from these examples suggest that these strategies are important best practices for WebAR in general, not just for specific brands.

## **2.8 Challenges in Implementing AR in E-Commerce**

While WebAR technologies have shown significant potential in addressing footwear-specific challenges such as virtual try-ons, size visualization, and reducing purchase uncertainty, several broader technical challenges still exist and must be considered when implementing AR in e-commerce platforms. In particular, the footwear industry faces additional complexity due to the need for accurate foot tracking, stable product visualization across different body types, and ensuring user comfort with interactive features in real-time environments.

In addition to these footwear-specific issues, the integration of AR technologies into e-commerce also encounters technical challenges related to performance optimization, 3D asset management, cross-browser compatibility, and network resilience. These technical limitations

apply across industries and are critical considerations for any WebAR application aiming for scalability and reliability in retail environments.

This section synthesizes relevant studies addressing these broader challenges and discusses potential solutions that can guide future AR implementations for both footwear e-commerce and other retail sectors.

### **2.8.1 Accuracy Problems and Real-Time Tracking**

Foot tracking within WebAR environments presents several technical challenges that stem from the limitations of current tracking technologies and the constraints of web-based platforms. One significant challenge is achieving accurate and real-time tracking of foot movements without the use of markers or specialized hardware. Visionbased tracking systems, while convenient, often struggle with occlusions and varying lighting conditions, which can impede the accurate detection of foot positions. For instance, [Foo et al. \(2022\)](#) developed a vision-based foot tracking algorithm that models the foot and shank using simple geometry. Their system, designed for overground gait assistive devices, demonstrated promising results but highlighted the difficulties in maintaining accuracy when the feet are partially occluded or when the camera's view is limited. Furthermore, [Bostanci et al. \(2013\)](#) reviewed various user tracking methods for augmented reality and emphasized the need for robust tracking techniques that can adapt to diverse environments and user movements.

### **2.8.2 Performance Optimization**

The performance in AR systems has a significant impact on user experience and its optimization is essential in AR applications. To improve user engagement and reduce technology fear, which could hinder consumer adoption, AR apps need to function seamlessly across a range of hardware specs (Dogra et al., 2023). According to research, poor performance can have a negative effect on perceived ease of use, which influences user satisfaction (Prasad et al., 2022). To guarantee that AR applications can manage heavy graphics loads while still being responsive and effective, thorough testing and optimization must be done. In order to lower latency and improve the viewing experience, which is connected to higher purchase intention, this involves improving the backend procedures ([Ong et al., 2019](#); Smink et al., 2019).

### **2.8.3 3D Asset Management**

One of the major challenges in AR or 3D-based websites is the complexity of producing and maintaining high-quality 3D models, which are essential to augmented reality experiences. E-commerce systems can offer immersive experiences without stressing system resources when 3D assets are managed effectively. Optimizing 3D models using methods like Level of Detail (LOD) is crucial since asset production must balance load speeds and visual fidelity ([Caboni & Hagberg, 2019](#)). According to research, in order to stay competitive and guarantee

a flawless augmented reality experience, businesses need to implement strategies that speed up the creation and implementation of 3D assets ([Khaldy et al., 2023](#); [Caboni & Hagberg, 2019](#)).

#### 2.8.4 Cross-Browser Compatibility

One of the remaining significant challenges in the AR domain is ensuring cross-browser functionality, particularly due to the inconsistent support for WebXR across different web browsers. To guarantee that AR capabilities are available to all possible users, web-based AR applications need to be adjusted to run on a variety of devices and operating systems ([Guo & Zhang, 2024](#); [Diana et al., 2023](#)). Because users may encounter constraints or frustrations depending on their browsing environment. Additionally, incompatibility can have a negative impact on sales and consumer adoption rates. To ensure a consistent experience and increase customer satisfaction, e-commerce platforms must integrate testing across various devices and browsers ([Kovács & Keresztes, 2024](#); [Yim et al., 2017](#)).

#### 2.8.5 Network Resilience

Another issue with AR in e-commerce is network resilience. Ensuring strong network performance is essential as AR apps depend more and more on cloud services ([McLean & Wilson, 2019](#)). Network infrastructure that can manage large data loads and render AR information in real time, even under less than ideal circumstances, is essential for e-commerce enterprises ([Javornik, 2016](#)). Resilient networks reduce latency and guarantee consistent access to augmented reality elements, improving the overall shopping experience and perhaps increasing retention rates ([Xu et al., 2024](#); [Wang et al., 2023](#)).

### 2.9 Addressing the Research Questions

**Table 2.3: Answering Research Questions**

| <i>Research Questions</i>                                                                                                     | <i>Answers</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b><i>How has web-based AR technology evolved in e-commerce applications from 2015 to 2025?</i></b>                           | In the five years from 2015 to 2025, e-commerce shift from high-barrier - desktop and app-based solutions to more approachable ones through WebAR. This was made possible by advancements in mobile hardware and the 2018 launch of WebXR, which allowed for AR experiences without the need to download apps. The problems still exist in terms of browser compatibility, foot tracking accuracy, and device performance (CatchMetrics, 2025; BrowserStack, 2025). |
| <b><i>What are the current technical challenge in implementing virtual try-on features for footwear?</i></b>                  | This question is already answered in Table 2.1 from Section 2.1.4 - Critical Reflection.                                                                                                                                                                                                                                                                                                                                                                            |
| <b><i>How does AR implementation influence user engagement, satisfaction, and purchase confidence in online shopping?</i></b> | AR implementation improves user engagement and satisfaction, along with confidence for purchasing in online shopping environments.<br><br>The unique immersive experiences, letting users interact with products and allowing them to try creates a deeper connection, and it usually increases the number of hours spent with the products.                                                                                                                        |

|                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                 | Secondly, realistic visualizations and virtual fittings provided by AR will allow consumers to make an accurate assessment, not perfect, of the fit and style of products, which contribute to increase the satisfaction level with their shopping experience.                                                                                                                                                                                                            |
| <b>How do various devices and browsers affect the performance, compatibility, and user experience of WebAR try-on features?</b> | As Liu et al. (2024) and CatchMetrics (2025) mention, WebAR performance and user experience are impacted in a negative way by low-end devices and limited browser support. Due to Safari's limited WebXR support compared to others, accessibility could be reduced.<br>According to studies, an increase of abandonment rates associated with load times above three seconds validates the importance of device capability and browser optimization in sustaining WebAR. |

## 3. Design

### 3.1 Introduction

The integration of augmented reality (AR) into e-commerce generates opportunities to enhance online shopping, particularly in the footwear domain since fit and appearance usually are the decisive factors to determine purchases. This project presents the creation of a web-based virtual shoe try-on platform that utilizes AR technology to enhance the interface between physical and digital shopping experiences.

Research shows consistently significant user preferences for virtual try-on systems. According to the comprehensive study of Dacko (2017) involving over 21,000 consumers, 87% attached great importance to accurate representation of the product, 82% value ease of use, while 71% preferred minimalist interface. These results mirror the research of Xue et al. (2019), Jiang et al. (2022) Singh (2024) which stresses the importance of intuitive navigation and simple design. In other work, both realism and ease of interaction have been indicated as critical success factors (Doe et al. 2021). Established user-centered design principles also support visual cognition ability, personalization, and visibility of functions (Norman, 1988; Gould & Lewis, 1985).

The project has a strategic technology stack revolving around WearFits, a commercial platform that allows not only virtual footwear visualization but also other accessories. The implementation balances technical capabilities with constraints to practical limit to test how AR shoe try-on works in real web-based shopping site. Examining the results from technical implementation and user experience related with virtual shoe try-on illustrates clearly how AR technologies can facilitate ecommerce interactions and affect consumer decision-making in the online footwear retail setup.

### 3.2 Design Choices

This section covers key design decisions that have been influenced through user research and best practices in augmented reality and e-commerce. Understanding user preferences is the heart of the success of any virtual try-on systems. Research have proven that realism and ease of interaction are important to users when using such technologies (Doe et al., 2021). An empirical study done by Dacko (2017) support these findings with statistics of 87% of users consider accurate product representation as the most important factor, while 82% give most priority to ease of use.

Moreover, minimalist interfaces where navigation is easy have more chances of increasing user satisfaction (Smith and Johnson, 2020), which support Dacko's finding, 71% of consumers prefer simple and clean interfaces. Clear visual cues and minimal steps to activate AR features were important for 67% and 74% of users, respectively.

### 3.3 Design Implementation

This section explains how the design choices were applied in practice using a user-centered design approach. This method prioritizes understanding user needs and preferences to make the interface simple and accessible. Wireframes, low-fidelity, and high-fidelity prototypes were created using Figma to visualize the website layout and user flow before development. The prototype is structured into three main sections:

A user-centered design approach is selected to design the website. This method emphasizes on understanding what online shoppers want the interface to be, which is a simple and accessible interface. The designs of wireframes and low-fidelity prototypes as well as high-fidelity were created using *Figma* to visualize the website layout and user flow before moving to the development stage. The prototype is structured into three main sections:

- *Homepage*: Products on display shelf, simulating real-world physical stores.
- *Product Page*: 3D shoe models that users can rotate, zoom, and view in detail.
- *AR Try-On Page*: AR try-on, with a QR code that launches the camera for trying on virtual shoes.

Simple navigation menus, large buttons, and clear instructions were designed to guide users throughout the process.

Several important factors were considered during the design phase:

- *Mobile Responsiveness*: Making sure the layout adapts well on different screen sizes.
- *Simplicity*: Using clean design, easy-to-read text, and recognizable icons.

- *Performance*: Optimizing 3D models for faster loading times.
- *Visual Engagement*: Maintaining a modern look and feel to enhance the shopping experience.

### 3.4 Final Design Overview

Building on user preferences, AR integration, and design principles discussed in earlier sections, the final product was developed with a strong focus on usability, simplicity, and engagement. Low-fidelity and high-fidelity prototypes were created using Figma. The following sections present these prototypes in detail, demonstrating the progression from conceptual layout to a fully realized design.

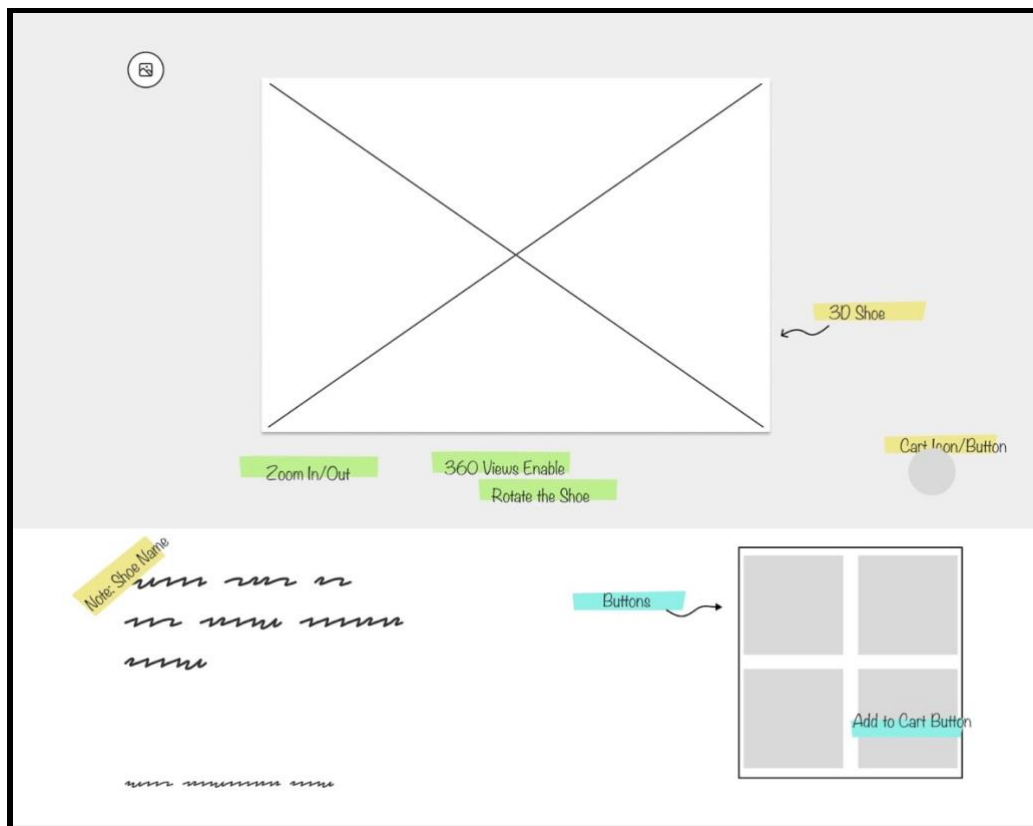


Figure 3.1: Low Fidelity Prototype of Product Page

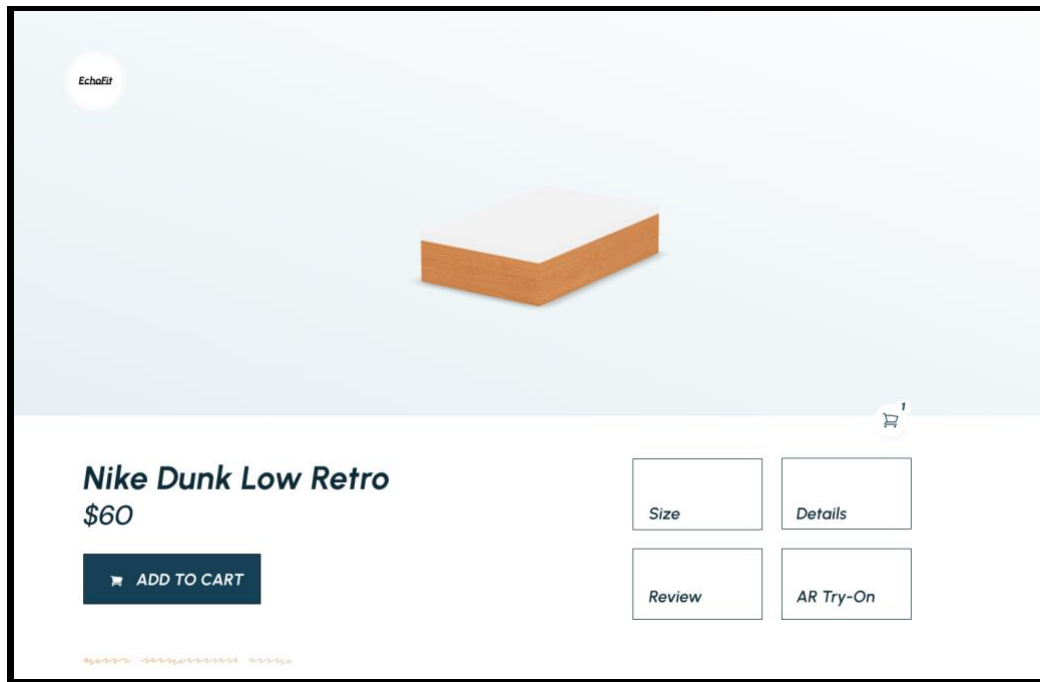


Figure 3.2: Low Fidelity Prototype of Product Page

### 3.5 Low-Fidelity Prototyping

The rest of low-fidelity prototyping is described in appendix 3 – Low Fidelity Prototype.

### 3.6 High-Fidelity Prototyping

The rest of high-fidelity prototyping is described in appendix 4 – High Fidelity Prototype.

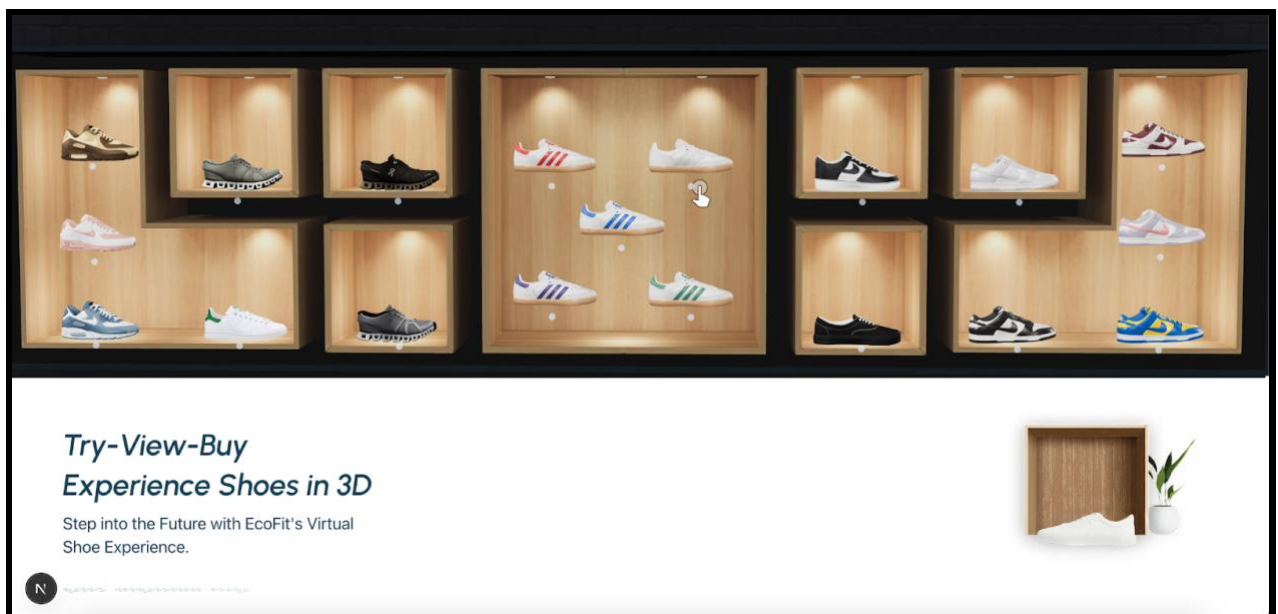
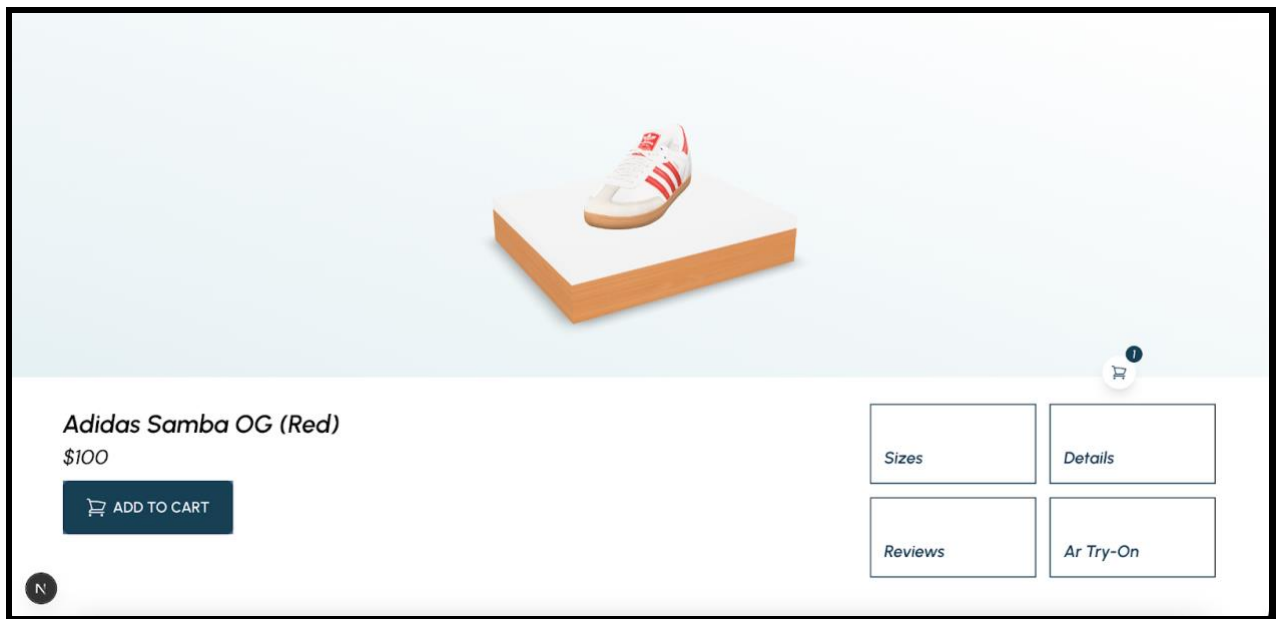


Figure 3.3: Actual Home Page Design



*Figure 3.4: Actual Product Page Design*

## 4. Implementation

### 4.1 Introduction

The implementation phase revolved around creating a functional Web-based Augmented Reality (WebAR) prototype that is intended to improve the online shoe shopping experience. Users can try on shoes virtually in real time using either their smartphones or any web camera-enabled devices and they do not need to download applications or install anything. Thus, it becomes a lightweight solution, accessible and perfectly suitable for users who browse through a standard mobile web browser.

The fundamental aim of the implementation is to enable real-time product visualization via browsers and allow users to view 3d shoes on their feet in their physical environment. This presents a more immersive and informative decisionmaking method for online shoppers, bridging the gap between physical and digital retail experiences.

This project's website is developed with NextJS, a React-based framework widely used in building modern front-end applications. There are two reasons why this framework is selected: it comes with built-in features (e.g. server-side rendering, routing, etc.) which reduce time-consuming from creating everything from scratch, and it has a large and active developer community, meaning it provides a large documentation, and tutorials. This makes it easier to troubleshoot problems and find solutions quickly during development. Because of these reasons, It is an ideal academic project with time limitations.

Unlike full-stack applications, this project does not require a backend server or database. The primary goal is to demonstrate web-based augmented reality (AR) and 3D visualizations for virtual shoe try-on. As a result, all content and functionality are managed on the front end. Using static assets and client-side scripts makes deployment simpler and ensures compatibility across devices and browsers without additional server-side logic or data storage. A complete breakdown of tools and technologies used in this project is shown in below table.

**Table 4.1: Technology Stack Overview**

| Category                         | Technology/Tools                        | Purpose                                                                                                                              |
|----------------------------------|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Front-End Framework              | Next.js                                 | Explained in above section                                                                                                           |
| 3D Visualization/<br>Interaction | Model Viewer                            | Lightweight, easy-to-use web component for embedding 3D models with AR support via WebXR. No complex setup required like Three.js.   |
| AR Try-On Platform               | Wearfits (Free Plan)                    | Enables AI-driven foot scanning and ARbased virtual shoe try-on. Used as a thirdparty tool due to project time/resource limitations. |
| 3D Model Format                  | glTF / glb                              | Efficient 3D file formats compatible with Model Viewer; optimized for web and AR experiences.                                        |
| Hosting Platform                 | Vercel                                  | Simple, scalable hosting for static Next.js sites with fast deployment and CDN distribution.                                         |
| Design Prototyping               | Figma (Low- Fidelity and High-Fidelity) | Used to create wireframes focusing on layout, navigation, and user flow before implementation.                                       |
| Version Control                  | Git + GitHub                            | Tracks code changes, supports collaboration, and maintains backup of project files.                                                  |

## 4.2 System Structure and Components

The prototype consists of the following main components:

### 4.2.1 Product Display Page

This area of the website contains a clean interface which showcases a selection of 3D shoe models. Each product is rendered interactively, allowing the user to rotate, zoom, and look at the shoe in 3D before engaging the AR feature.

### 4.2.2 Shoe Selection and UI Interaction

Users can browse and choose from a few shoes using a simple yet self-explanatory interface. Options are presented via cards or thumbnails linked with its own 3D version. A button is dedicated for each element - "AR Try-On" to push people into the AR session.

### 4.2.3 WebAR Try-On Viewer

When the "AR Try On" button is pressed, the QR code will appear and scanning it will bring the WearFit WebAR viewer launches and activate the user's camera. The selected 3D shoe model is placed on the user's foot in real time using markerless tracking. The projection responds to user movement and perspective thus allowing for a realistic and interactive AR try on experience.

### 4.2.4 Device Compatibility

The experience is directly accessible through most modern browsers on desktop and mobile devices (for example, Chrome, Safari, Edge) without needing to install additional software for the experience. Dynamic resizing based on various screen and light conditions while maintaining alignment and realism happens in the AR viewer.

### 4.2.5 AR Integration

However, this project will implement the AR try-on functionality through Wearfits, a commercial solution developed specifically for the visualization of virtual footwear. Wearfits has a free version which offers an accessible option for academic research. While the free plan comes with all the features of AR and AI-based foot scanning, it has the limitation that 3D shoe models uploaded into the platform automatically delete after six days. However, this limitation does not make Wearfits an unsuitable solution for the present project, as it is quick, real-time accessibility for accurate virtual fitting cover up for its drawback.

Developing a fully custom AR try-on system from scratch—especially one that includes real-time foot tracking, 3D rendering, and cross-platform support—is highly complex and beyond the scope of this dissertation's time and resource constraints. Therefore, using a third-party solution like Wearfits is a strategic and time-efficient decision.

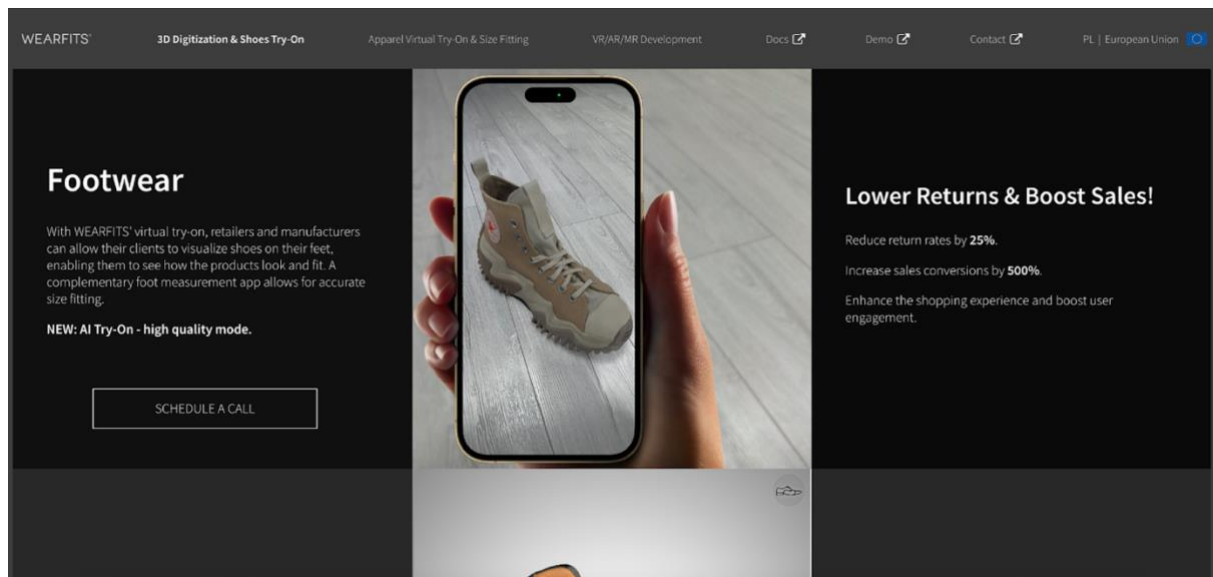
While there are other commercial platforms offering similar AR try-on features— such as **Vyking, Wanna Kicks, and Modiface**—these typically require paid licenses or partnerships, which are not feasible within the context of this academic project. Thus, Wearfits stands out as a viable, no-cost solution that enables the implementation and evaluation of AR try-on functionality within a realistic web-based retail scenario.

### 4.3 WearFit

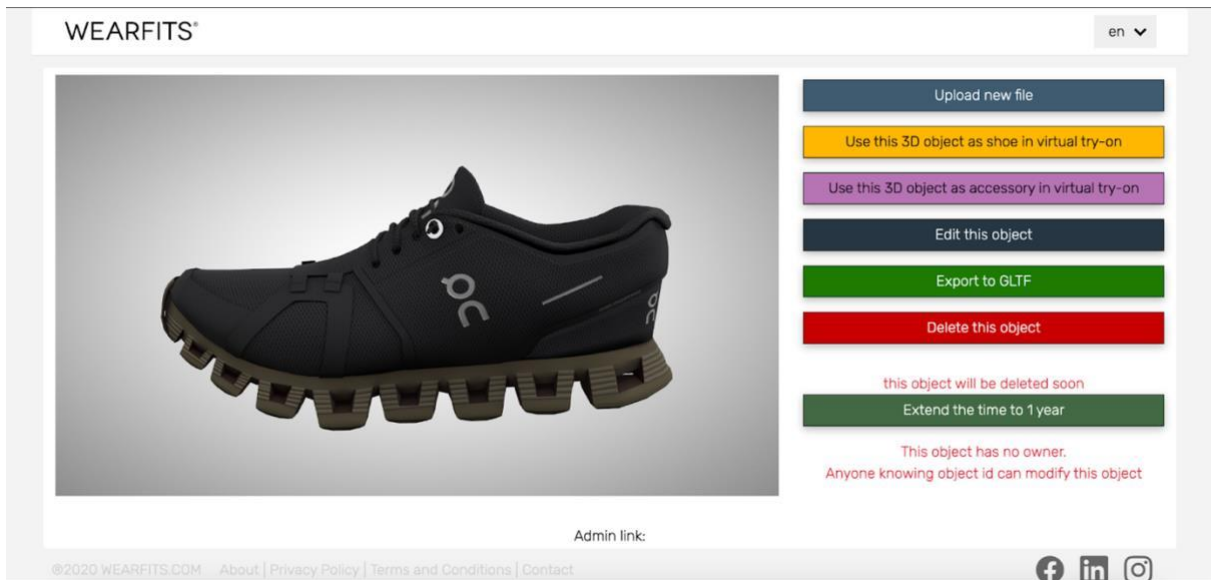
Wearfits is a commercial platform that enhances the online fashion shopping experience by combining augmented reality (AR) and artificial intelligence (AI) technologies. While it is popularly known for allowing users to try on shoes virtually in real time, the platform has expanded to include other types of apparel such as clothing and accessories.

The system operates by allowing a customer to visualize wearing different fashion items while being present in his or her real ambiance on the smartphone or computer with a webcam. Hence, the offline and online retail is being connected, making online shopping interactive, engaging, and more personal.

Further, the platform also has markerless AR tracking functionality, meaning users can use the platform without added printed markers or special equipment. Brands and retailers can attach Wearfits neatly with their current e-commerce platforms, and thus it is fit for the fashion industry, footwear or not, in terms of flexibility and scalability.



*Figure 4.1: Official Website of WearFit*



*Figure 4.2: WearFit's Virtual Try Experience Admin Panel*

#### 4.4 3D Visualization

For 3D visualization, this project uses Model Viewer, rather than more complex frameworks like Three.js. Model Viewer is a lightweight, open-source web component developed by Google that allows developers to embed and interact with 3D models using a simple HTML tag. It supports modern features such as AR integration (via WebXR), responsive scaling, camera controls, and fast model loading, all with minimal code. Unlike Three.js—which requires significant setup and manual coding of 3D scenes—Model Viewer provides an easier and more accessible solution for embedding high-quality 3D models, making it ideal for front-end-focused projects with limited time and scope.

According to Laksono and Aditya (2019), fast-loading, responsive 3D tools are essential for user retention, especially on mobile devices. Model Viewer meets this requirement by supporting efficient formats like glTF and USDZ, ensuring optimal performance across platforms. Its built-in AR support eliminates the need for custom WebXR integration, allowing seamless transitions between 3D viewing and AR try-on modes.

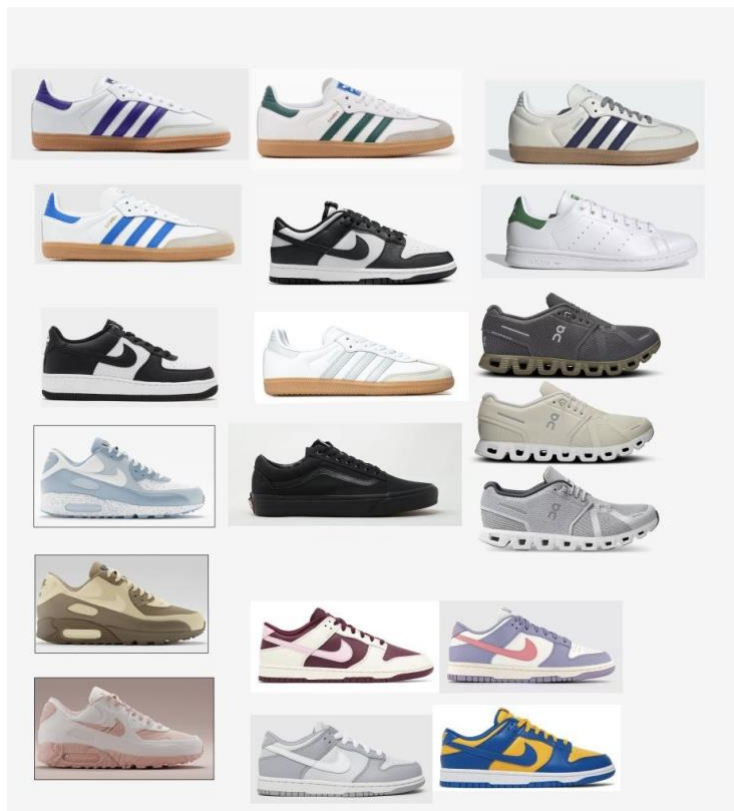
In summary, the decision to use Next.js with Model Viewer, without a backend or database, was based on the project's focus on speed, simplicity, and AR/3D visual delivery. This architecture is well-suited to the goals of demonstrating immersive tryon experiences while maintaining technical feasibility within the project's time constraints.

#### 4.5 3D Shoes

The 3D shoe models used in the project were sourced from **CGTrader**, a commercial 3D model marketplace. A selection of ten shoe models was purchased to represent a diverse range

of styles including athletic, casual, and formal footwear. Each model was chosen based on polygon optimization (averaging 30,000-50,000 polygons per model) to ensure smooth performance across various devices while maintaining visual fidelity. The models came with pre-applied textures at 1024x1024 resolution, providing realistic material representation without requiring extensive modification.

The website includes eighteen different shoe models from CGTrader, chosen to show how the AR try-on feature works with various styles. The shoes were selected for their visual variety and technical differences, with varying levels of detail, materials, and structures. This diverse selection helps test how well the AR system can display different types of footwear, from simple to complex designs. The shoes presented in the website are illustrated in below figure.



*Figure 4.3: Total Shoes Used on the Website*

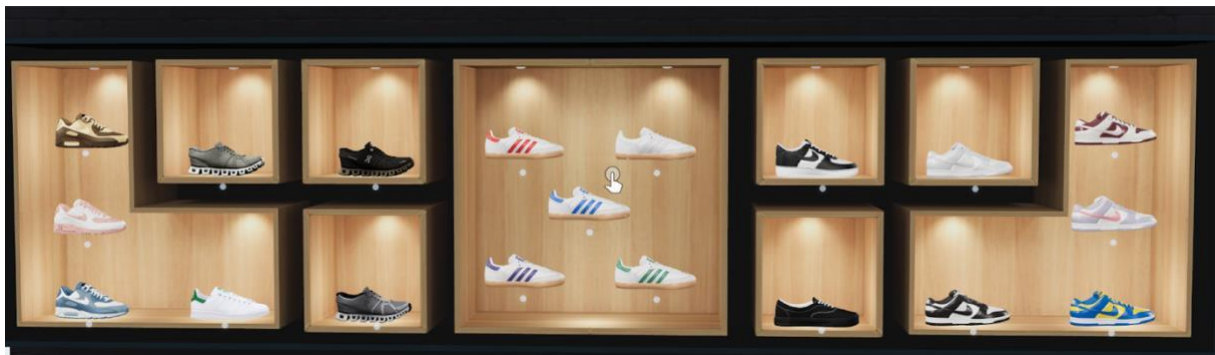
The 3D models were converted from their native formats to glTF (GL Transmission Format) to optimize web delivery and ensure compatibility with WebXR standards for the AR implementation. This format choice supported the project's focus on minimizing loading times while preserving essential visual details like stitching, texture, and sole patterns that contribute to realistic product visualization.

#### **4.6 3D Display Shoe Shelf**

The 3D Shoe Shelf is the main entry point to the e-commerce website, designed to feel like a real-life shoe store. It features a dark background with wooden compartments arranged in a neat grid, displaying around 20 athletic and casual shoes.

As the biggest 3D element in the project, the shelf was carefully optimized to keep it smooth and fast without losing visual quality. Each shoe is displayed with realistic textures, lighting, and size, supporting Dacko's (2017) finding that users care most about accurate product images.

The shelf is also interactive users can click on shoes using selection dots, making it easy to explore. This matches research by Xue et al. (2019), Jiang et al. (2022), and Singh (2024), who stress the importance of simple, intuitive navigation. Overall, the design offers a more engaging and familiar shopping experience while keeping a minimalist layout, which 71% of users prefer (Dacko, 2017).



*Figure 4.4: 3D Shoe Display Shelf*

This implementation confirms that WebAR can offer a convenient and engaging way to enhance the online footwear shopping experience. While the system functioned well on most modern devices, limitations related to accuracy, browser support, and device performance remain. Future improvements could involve incorporating AI-based foot detection and enhancing fit personalization to provide a more accurate and helpful try-on experience.

## 4.8 Virtual Try-On

To integrate the virtual try-on feature in the proposed AR web platform, WearFits' the "Try-On" feature is used. WearFits is a commercial platform that helps brands offer virtual try-on experiences. In addition to shoes, WearFits allows users to try on apparel and spectacles in virtual fitting rooms. However, the "Try-On" features used in this project is free version and multiple downsides are uploaded 3d shoe model will be deleted after six days and other paid features. This website included a straightforward WebAR experience that allows users to view the pair on their feet using a mobile device's camera without downloading an app as shown in below figure.

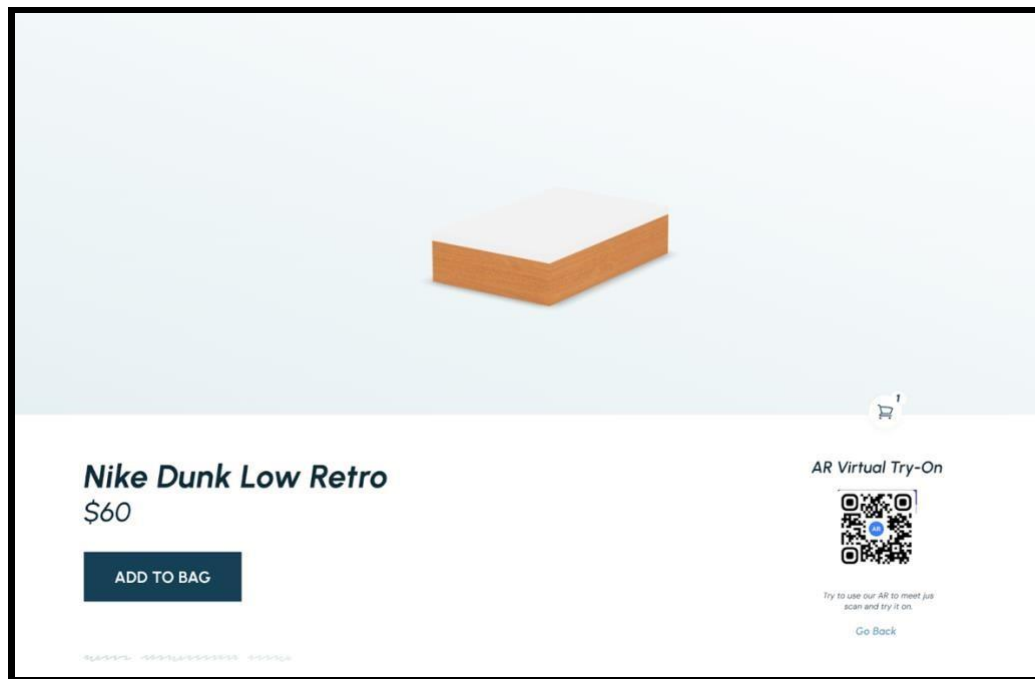


Figure 4 - 5: Virtual Try-On QR Code



*Figure 4.6: Virtual Shoe Try-On Experience*

By scanning (pointing) the QR code on the product page, the user's camera is opened during the try-on process, allowing the chosen shoe model to be added over the prepared foot in real-time mode which can be seen in figure . A sample of the WearFits virtual try-on interface shown in figure 5.6, which provides insight into the user experience during the trial period. It allowed users to rotate, move, and adjust the shoe in real time, helping them feel more confident about their purchase. Overall, using WearFits helped test how well WebAR technology can support online shoe shopping and made the virtual experience feel closer to shopping in a real store.

## 5. Evaluation

To evaluate and test the proposed website, the evaluation processes include the usability, performance, and effectiveness of the footwear try-on system. This involved real real-world based testing simulating and users interacting with the prototype across various devices and different interned speed. This way, valuable insights can be gained on what are the strengths and what requires improvement, particularly in aspects of interface clarity, AR functionality, and responsiveness.

The evaluation considers quantitative metrics, such as task completion rates, time taken for key activities, and user satisfaction ratings, alongside qualitative feedback to identify the areas need to be improved. The results of the evaluation produce insight on how the system can be improved in the future to better enhance the online shopping experience using augmented reality.

According to StatCounter Global Stats for March 2025, the cross-browser test for the project was conducted on Google Chrome (69.44% share), Safari (18.32%), Microsoft Edge (5.27%), and Mozilla Firefox (3.19%). This selection ensures that about 96% of the user base among desktop and laptop users around the world used Chrome, Safari, Edge, and Firefox mostly. All the testing for the website, 3D Models and AR will be conducted on these browsers.

**Table 5.1: Browser-Based Testing Method**

| <b><i>Tool/Method</i></b> | <b><i>What it does</i></b>                     |
|---------------------------|------------------------------------------------|
| <b>Manual testing</b>     | Open a site on each browser (desktop + mobile) |

This section presents the evaluation of rendering performance of the 3D shoe model used in the project on the four browsers. Depending on browsers' rendering engines and graphical support, the performance including load times can be different. The aim of this test is to ensure the model displays correctly, load quickly and performs smoothly across the chosen browsers which are Chrome, Safari, Firefox, and Edge. The figure 6.1 describes what tests will be conducted on the 3d contents.

| <b>Metric</b>           | <b>Tools Used</b>                                     | <b>Pass Criteria</b>                      | <b>References</b>                                    |
|-------------------------|-------------------------------------------------------|-------------------------------------------|------------------------------------------------------|
| <i>Loading Time</i>     | <i>performance.now ()</i><br><i>Chrome DevTools</i>   | <i>under 3 seconds</i>                    | <i>BrowserStack</i>                                  |
| <i>Modal File Size</i>  | <i>File Inspector, DevTools</i>                       | <i>Under 5MB and preferably under 2MB</i> | <i>Optimizing Media and File Size (Google)</i>       |
| <i>Frame Rate (FPS)</i> | <i>stats.js, performance.now()</i>                    | <i>Consistent between 55-60 fps</i>       | <i>Rendering Performance Best Practices (Google)</i> |
| <i>GPU Memory</i>       | <i>Spiker</i>                                         | <i>No Major Memory Usage</i>              | <i>Memory Overview &amp; Usage (Chrome DevTools)</i> |
| <i>Lighthouse Score</i> | <i>Google Lighthouse Tool</i>                         | <i>Performance score over 80</i>          | <i>Lighthouse Performance Scoring (web.dev)</i>      |
| <i>Memory Usage</i>     | <i>Chrome Manager,</i><br><i>performance.memory()</i> | <i>Under 200 MB</i>                       | <i>(Khronos Group, n.d.)</i>                         |

*Figure 5.1: Performance Targets and Tools for Evaluating Web-Based Virtual Try-On Applications*

## 5.1 Shoe Display Shelf Model

### 5.1.2 Desktop Testing

All testing focuses exclusively on the 3D shoe display shelf model rendered on desktop devices. This includes evaluations of average loading times, test environment specifications, and estimated loading durations under varying internet speeds.

#### Loading Time

**Table 5.2: Testing Condition Specifications**

| <b>Test Condition</b> | <b>Details</b>                             |
|-----------------------|--------------------------------------------|
| Internet Speed        | 96 Mbps (Wi-Fi)                            |
| Browser Cache         | Disabled                                   |
| Device                | Macbook Pro M1 (Desktop: high-performance) |
| 3d Model File Size    | 76.1 MB                                    |

**Table 5 - 3: Results of Loading Time on Desktop Devices**

| <b>Browser</b> | <b>Average Load Time (ms/s)</b> | <b>Remarks</b>              |
|----------------|---------------------------------|-----------------------------|
| Chrome         | 3051.8 ms (3.05s)               | Fast and consistent         |
| Firefox        | 3090.9 ms (3.09s)               | Close to Chrome             |
| Edge           | 3143.8 ms (3.14s)               | Slightly slower than Chrome |
| Safari         | 4344.4 ms (4.34s)               | Significantly slower        |

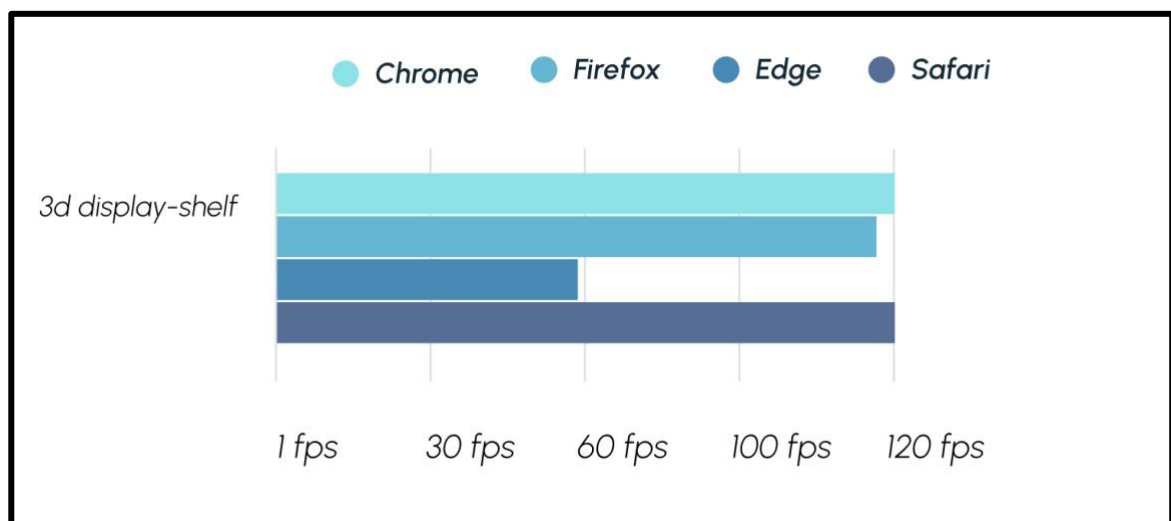
A benchmark test evaluated on various browser and its capabilities in loading 3D models on a desktop device with a 96 Mbps Wi-Fi network and browser cache disabled, simulating a first-time user experience. Among tested browsers, Chrome stands out as the fastest among browsers with an average load time of around *3052 ms*, while Firefox and Edge showed similar good speeds. With load times of more than *4000 ms* on average. Safari is the worst among other browsers. Such clear latency can be attributed to the very limited support provided by Safari for WebGL and WebXR features for rendering 3D models. Hence, Safari renders the 3d model on a more constricted graphics engine with no native support of WebXR, which again plays a big role in loading speed and rendering performance.

Chrome, meanwhile, is well-optimized for 3D web experiences with excellent support for WebGL, GPU acceleration, and fast JavaScript execution. Overall, Chrome, Firefox, and Edge were the best-performing 3D model loading browsers under this test setup.

### Performance or FPS

For augmented reality applications, a smooth user experience requires stable frame rates. Official Google documentation indicates that 30 fps is minimal, while 60 fps is ideal for smooth interaction. 55 fps is the average stable frame rate that handles effective data streaming with a reduced CPU load for a true AR experience.

Studies show that websites with performance scores of 80 or higher keep users on the site longer and encourage them to interact more. Schmidt and colleagues found clear evidence of this in 2019 (Nienhueser et al., 2015). This matters most for online shopping websites like Tokopedia and Shopee, where better website performance leads directly to more sales. When shopping sites run smoothly and quickly, customers are more likely to complete their purchases rather than leave out of frustration (Muna et al., 2022).



*Figure 5.2: Average FPS Across Browsers*

The 3D model performance report across different browsers reveals notable differences in rendering efficiency and consistency. Brave emerges as the top performer, delivering a perfect 120 FPS. This indicates a smooth and highly stable experience with minimal lag. Chrome and Edge also performed very well, maintaining 120 FPS during record times. In contrast, Safari shows the weakest performance across almost all metrics. It struggles to maintain smooth visuals with an average FPS of only 59.4, nearly half of what the other browsers achieved.

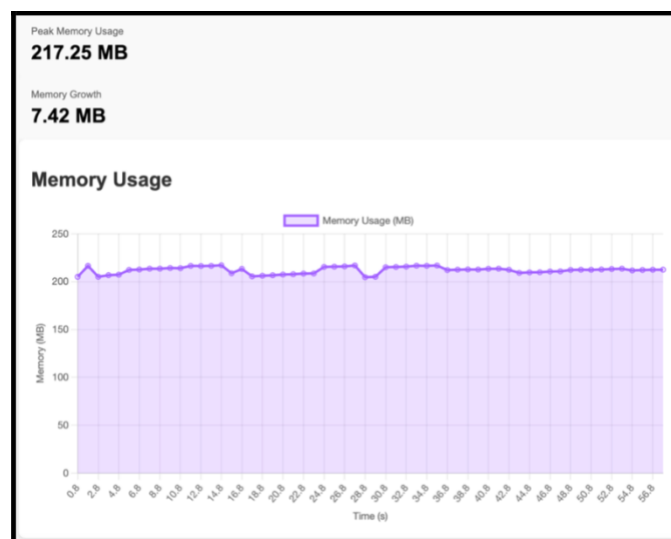
Overall, Chrome stands out as the most optimized browser for handling interactive 3D content, closely followed by Firefox and Edge. Safari, on the other hand, may not be suitable

for performance-critical 3D applications due to its inconsistent frame delivery and lower responsiveness.

## Memory Allocation

In a 3D web app, memory usage would generally be of two main types: GPU memory and Javascript heap memory. GPU memory or VRAM is the part of memory that is mostly used for storing assets required for rendering like 3D mesh data, textures, materials, and shaders; therefore, it comes in handy when you are using graphics card or WebGL engine to ensure smooth visual performance. In turn, JavaScript heap memory is managed by the browser and stores runtime logic and data structures of the application, parsed model files, and components of interfaces. They both have separate activities, but it is just as important to keep a tab on either type because very high memory usage in any of these might affect performance, especially on low-resource devices.

## Chrome Browser (Desktop)



*Figure 5.3: Memory Usage on 3d Display Model*

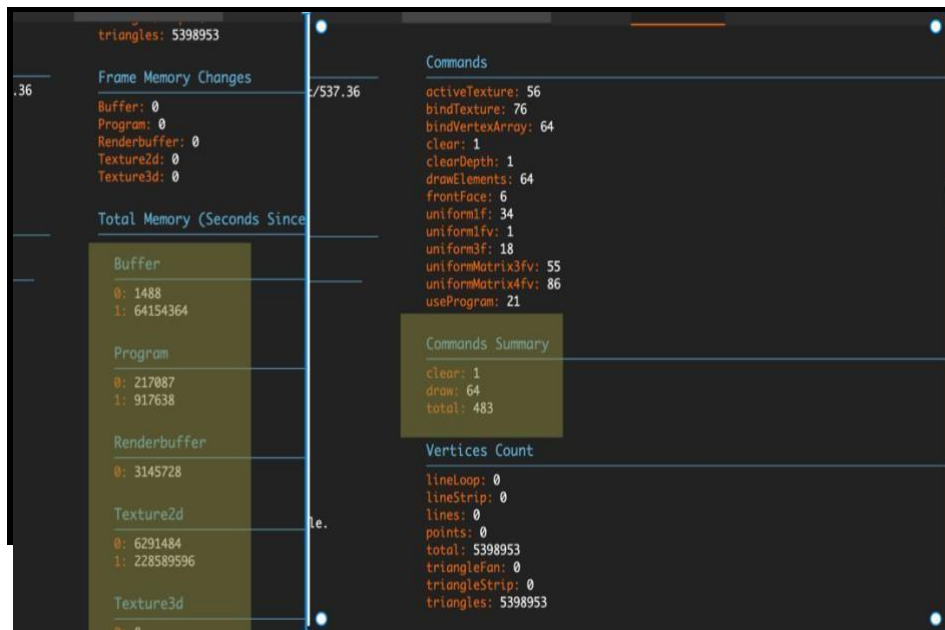


Figure 5.4: Spiner Web Plugin for GPU Memory Usage

Table 5.4: JavaScript Heap Usage Overview

| Filename               | Peak Usage | Memory Growth |
|------------------------|------------|---------------|
| 3d display shelf model | 217.25 MB  | 7.42 MB       |

Table 5.5: GPU Memory Usage Analysis (Excludes JS Heap)

| Filename               | Estimated GPU Usage |
|------------------------|---------------------|
| 3d display shelf model | 300 - 320 MB        |

An analysis of the 3D Shoe Display Shelf scene was conducted using WebGL inspection tools such as Spector.js. The findings show that the rendering process involves approximately 64 draw calls and handles a total of 5.39 million triangles, indicating a high level of geometric complexity.

The GPU memory usage was estimated at around 303 MB, broken down into approximately 64 MB for geometry buffers and 235 MB for textures, with the largest texture consuming over 228 MB alone. These values reflect the resources allocated specifically to the GPU (VRAM), such as mesh data, textures, shaders, and rendering buffers. It's important to note that this measurement does not include JavaScript heap memory, which stores the logic,

UI components, and initial model parsing. When combined with heap usage, the total memory impact of the application is considerably higher.

Testing memory and graphical performance in **Firefox** and **Safari** presents certain limitations compared to Chrome-based browsers. Unlike Chrome and Edge, which offer a built-in Task Manager with detailed per-tab memory usage and GPU consumption, Firefox's tools such as about: performance and about: memory do not provide a straightforward graphical overview or real-time GPU memory statistics. Similarly, Safari lacks detailed built-in developer tools for monitoring GPU memory usage and draw call performance. As a result, gathering accurate data on JavaScript heap usage, GPU load, and rendering metrics in these browsers often requires external profiling tools or more complex workflows. This makes it more difficult to directly compare 3D web performance across all major browsers, especially when targeting optimization for lower-end systems or mobile users.

### 5.1.3 Mobile Testing

#### Loading Time

This section evaluates and compares average loading times for 3D models with 3G and 4G network types based on performance tests using Chrome, Firefox, Safari, and Edge under various simulated internet speeds.

Since network simulation profiles differ by browsers, there are some blank spots in the table. For example, Chrome has specific profiles like "Slow 4G" and "Fast 4G," defined by states of bandwidth and latency, while Firefox has only one generalized profile set for "4G," at some 4 Mbps and about 20 ms latency. Because of the lack of the same measurement settings, it is difficult to perform one-to-one comparisons across all 4G levels in Firefox and other browsers.

**Table 5 - 6: Mobile Testing Condition Specifications**

| <b>Test Condition</b> | <b>Details</b>               |
|-----------------------|------------------------------|
| Internet Speed        | 3G,4G                        |
| Browser Cache         | Disabled                     |
| Device                | Mobile Simulation on Browser |
| 3d Model File Size    | 76.1 MB                      |

**Table 5 - 7: Summary of average loading time on mobile devices**

| <b>Internet Speed</b>      | <b>Chrome</b> | <b>Firefox</b> | <b>Safari</b> | <b>Edge</b> | <b>Average Time</b>    |
|----------------------------|---------------|----------------|---------------|-------------|------------------------|
| 3G (750kbps)               |               | 8.26 min       |               |             | 50.4 min               |
| 3G (380Kbps)               | 50.4 min      |                |               | 1.11 hours  |                        |
| Slow 4G<br>(400 Kbps)      | 6.66min       |                |               | 7.9min      | 5.34 min               |
| Fast 4G<br>(4 Mbps)        |               | 1.52 min       |               | 47.4s       | 1.52 min               |
| Regular 4G<br>(1.6 Mbps)   | 2.42 min      |                |               | 2.6 min     | 2.42 min               |
| Fastest 4G<br>(14–16 Mbps) | 24s           | 26s            | 24s           | 26s         | 0.415 min<br>(~25 sec) |

**Table 5 - 8: Average loading time on mobile devices with 3G and 4G internet speeds**

| <b>Internet Speed</b> | <b>Average Time</b> |
|-----------------------|---------------------|
| 3G                    | 29.33 minutes       |
| 4G                    | 2.42 minutes        |

Loading times for 3D models were summarized under different internet speeds, particularly focusing on 3G and 4G network types. The testing was undertaken under simulated conditions of four major browsers such as Chrome, Firefox, Safari, and Edge to mimic real-world scenarios faced by users. Results indicated that internet speed greatly affects performance.

Overall, the findings emphasize the importance of optimizing 3D web content not only for high-speed connections but also for varying browser behaviors and slower networks. By understanding these differences, developers can make decisions to ensure accessible and efficient WebAR and 3D experiences for all users.

## Performance or FPS

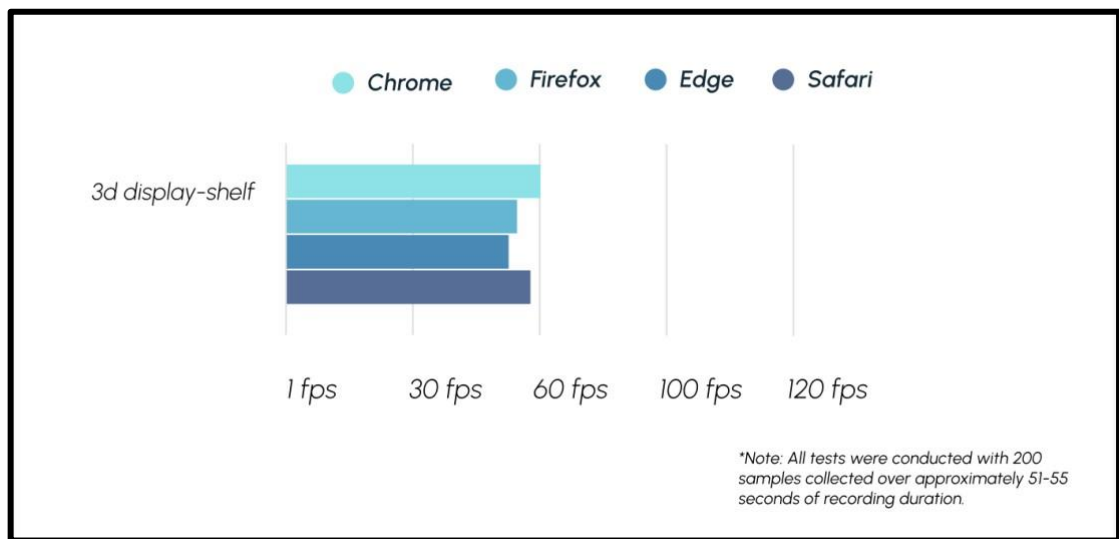


Figure 5.5: 3D Model Performance Comparison Across Mobile Browsers

According to the reports on performance of 3D modeling, the mobile browsers are rendering frames at an acceptable rate. Most of them operate with frame rates beyond the threshold of 30 FPS, which is considered the minimum limit required for an interactive 3D scene. The leading mobile browser is Chrome Mobile, which has a frame rendering speed of 60.0 FPS, an ideal frame per second rate for most displays and Safari is in second place with 57.3 FPS, followed by Firefox Mobile, while Edge Mobile still does well at 54.1 FPS. Nevertheless, there is no significant notable different among four browsers and all has a consistent user experience across major mobile platforms, with Chrome slightly ahead in optimization.

## 5.2 Shoe Models

### 5.2.1 Desktop Testing

All testing focuses exclusively on the 3D shoe models rendered on desktop devices. All testing configurations are the same as above desktop setup including evaluations of average loading times, test environment specifications, and estimated loading durations under varying internet speeds.

### Loading Time

The total amount of files used in the project can be seen in the table below. Due to the vast numbers of files for 3D shoe models and time constraints in evaluation with manual testing, it was decided to go for a representative sampling method. Models were grouped under brand categories such as Adidas, Nike, Cloud-9, and Vans, and one model from each brand was selected based on either its complexity of the file or for its size. Priority was given to larger files, as they are more likely to encounter performance or loading challenges. Hence, the worst-case scenario for testing was created.

The final selection included:

- Adidas Stan Smith (9 MB) as the largest Adidas model,
- Cloud-9 Snow (8.3 MB) for its relatively high file size among the Cloud-9 group,
- Nike Air Force One (11.2 MB) as the largest and most graphically rich file,
- Vans Black (5.3 MB), the only model which has average file size among them, and
- Cloud-9 Sand (2.4 MB), since it is the smallest one.

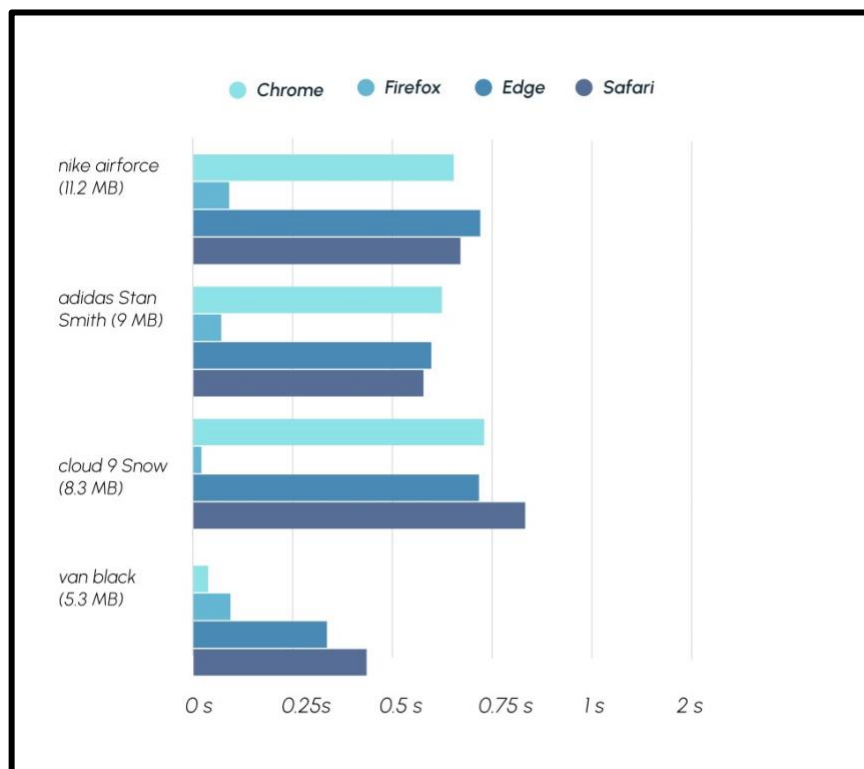
This set of five models represents the brands while maintaining testing efficiency. The results obtained from these samples provide reliable insights into the general performance cross-section of the entire library of 3D models.

**Table 5.9: Test Condition Specifications**

| <b>Test Condition</b> | <b>Details</b>                                                                                                                   |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Internet Speed        | 96 Mbps (Wi-Fi)                                                                                                                  |
| Browser Cache         | Disabled                                                                                                                         |
| Device                | Macbook Pro M1 (Desktop: high-performance)                                                                                       |
| 3d Model File Size    | Adidas Stan Smith (9 MB)<br>Cloud-9 Snow (8.3 MB)<br>Nike Air Force One (11.2 MB)<br>Vans Black (5.3 MB)<br>Cloud-9 San (2.4 MB) |

**Table 5.10: Results of Loading Time Across Browsers on Desktop Devices**

| <i>Model</i>             | <i>Size</i> | <i>Chrome</i> | <i>Firefox</i> | <i>Edge</i> | <i>Safari</i> |
|--------------------------|-------------|---------------|----------------|-------------|---------------|
| <i>nike-air force-on</i> | 11.2 MB     | 0.695s        | 0.112s         | 0.73s       | 0.707s        |
| <i>adidassam-smith</i>   | 9 MB        | 0.673s        | 0.104s         | 0.615s      | 0.603s        |
| <i>cloud-9snow</i>       | 8.3 MB      | 0.77 s        | 0.024s         | 0.716s      | 0.811s        |
| <i>van-black</i>         | 5.3 MB      | 0.051s        | 0.099s         | 0.382s      | 0.431s        |



**Figure 5 - 6: Loading Time Across Browsers on Desktop Devices**

Loading time evaluation across four browsers showed notable differences in 3d model loading times. Although Chrome consistently performed well, particularly with lighter models such as van-black (0.051s), its performance against heavier models such as nike-air-force-one was slower (0.695s) which is still acceptable. Firefox has a faster overall loading times, averaging 0.024s for cloud 9 and 0.104s for adidas sam smith. On the other hand, Edge has shown steady performance but slightly slower than those of Firefox through all models. In contrast, Safari always has consistently shown the slowest load time which indicates less optimization for WebGL-based model rendering.

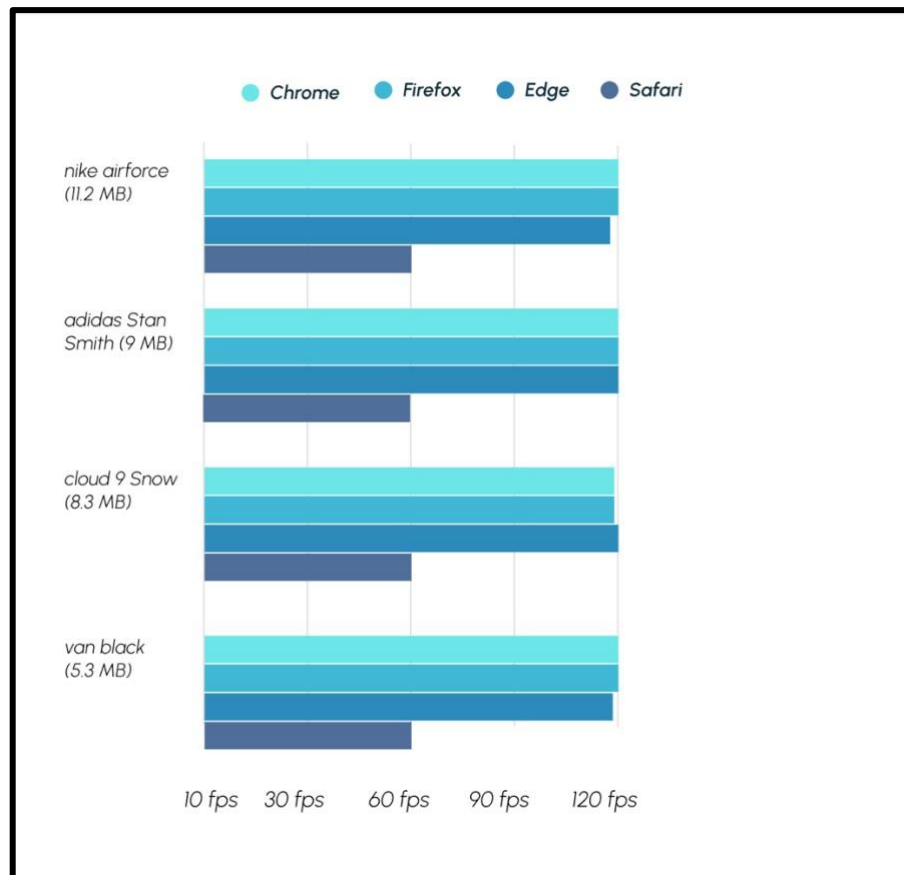
All these results highlight the importance of cross-browser testing because loading times of models differ depending on the size of the model and capabilities of browsers, which eventually make a huge difference to the user experience in WebAR applications since the load time across browsers varies on how each handles WebGL rendering, JavaScript execution, threads prioritization, and resource caching.

### Performance or FPS

This section of report presents a comparative analysis of the frame-per-second (FPS) performance when rendering four 3D models across major desktop browsers: Chrome, Firefox, Safari, and Microsoft Edge. The models tested include: *nike-air force-one-plate.glb*, *adidas-sam-smith*, *cloud-9-snow-plate.glb*, and *van-black-plate.glb*. Each model was tested under identical conditions on a high-performance desktop environment.

**Table 5.11: Summary table comparing the performance metrics from each browser's 3D Model on Desktop Devices**

| <b>Metric</b>                | <b>Chrome</b> | <b>Firefox</b> | <b>Safari</b> | <b>Edge</b> |
|------------------------------|---------------|----------------|---------------|-------------|
| nike-air force-one-plate.glb | 120fps        | 120fps         | 60fps         | 119.4fps    |
| adidas-sam-smith             | 120fps        | 120fps         | 60fps         | 120fps      |
| cloud-9-snow-plate.glb       | 119fps        | 119fps         | 60fps         | 120fps      |
| van-black-plate.glb          | 120fps        | 120fps         | 60fps         | 119.7fps    |



*Figure 5 - 7: Average FPS Across Browsers*

All three Chromium-based (i.e. Chrome and Edge) and Firefox browsers showed nearly or fully 120 FPS on all tested models. As it is clear, the WebGL engine and utilization of the underlying GPU across platforms are highly effective. The 3D models used within the project are clearly optimized for modern browser performance optimization. Out of all the browsers, Safari performance is capped at 60 fps which is exactly half of the performance seen in others. Nevertheless, FPS performance hardly shows variation between models in FPS, implying the model complexity is wellmanaged and does not impact rendering speed on desktop hardware. Considering the identical results from all browsers further supports that these 3D models are optimized for real-time rendering.

Overall, the results demonstrate that the tested 3D models are highly performant on all major desktop browsers, with the exception of Safari's FPS ceiling. Developers targeting high-refresh-rate displays may need to account for this behavior in Safari, but general performance remains solid. The consistent 120 FPS rates on Chrome, Firefox, and Edge confirm strong rendering stability and efficient use of system resources.

## Memory Allocation

The evaluation criteria and structure for 3d models' memory usage are the same with desktop step up which is already discussed in above *Memory Allocation under 5.2.1 Shoe Display Shelf Model*.

### Chrome Browser (Desktop)

**Table 5 - 12: JavaScript Heap Usage Overview**

| <b>Filename</b>              | <b>Peak Usage</b> | <b>Memory Growth</b> |
|------------------------------|-------------------|----------------------|
| nike-air force-one-plate.glb | 34.96 MB          | 0.84 MB              |
| adidas-sam-smith             | 57.42 MB          | 1.60 MB              |
| cloud-9-snow-plate.glb       | 78.39 MB          | -0.68 MB             |
| van-black-plate.glb          | 24.78 MB          | 1.99 MB              |

**Table 5 - 13: GPU Memory Usage Analysis (Excludes JS Heap)**

| <b>Filename</b>              | <b>Estimated GPU Usage</b> |
|------------------------------|----------------------------|
| nike-air force-one-plate.glb | 54.75 MB                   |
| adidas-sam-smith             | 2.77 GB                    |
| cloud-9-snow-plate.glb       | 157.8 MB                   |
| van-black-plate.glb          | 63 MB                      |

According to the results from the above table, all four models demonstrate low heap consumption in JavaScript. Its peak use was noted at 78.39 MB, with cloud-9-snow-plate.glb at the highest, while the lowest and the most efficient vanblack-plate.glb, used only 24.78 MB. In addition, across all models, memory growth was relatively small, with cloud-9 even having negative growth (-0.68 MB), suggesting that garbage collection or memory release has been done effectively over time.

In terms of graphical usage, the most standout model is the adidas-stan-smith model and its memory usage (2.77 GB) is greatly different from the other models' values. It means that a large GPU footprint is likely due to heavy or uncompressed texture assets. In contrast, the nike-air-force-one-plate.glb even needs only 54.75MB of GPU memory, revealing that optimized texture can keep overall usage low. However, GPU memory consumption varies widely from 54 MB optimized by nike-air-force-one to 2.77 GB unoptimized by adidas-stan-smith. This 50x difference underscores how much texture optimization affects 3D web performance even if the efficiency of the JavaScript code remains the same. In conclusion, the memory profiling shows that the 3D model is using JavaScript heaps efficiently (24-78 MB) and is growing very little. Memory analysis reveals efficient JavaScript heap usage (24-78 MB) with minimal growth across all 3D models.

To conclude, memory analysis shows that for the 3D models, the JavaScript heap is being efficiently used (24-78 MB) with very little growth. The difference in GPU memory consumption shows the other side of the story, where nike-air-force-one's uses 54 MB while adidas-stan-smith's 2.77 GB. This nearly 50-times difference is a strong indicator of how important texture optimization is for performance in 3D web applications. Without it, even well-written JavaScript cannot save the experience from becoming slow or unstable.

## 5.2.2 Mobile Testing

### Loading Time

This section evaluates and compares average loading times for four 3d shoe models, which are selected to provide ideas of how long it takes to take in the browsers, with 3G and 4G network types based on performance tests using Chrome, Firefox, Safari, and Edge under various simulated internet speeds.

**Table 5 - 14: Test Condition Specifications**

| <b><i>Test Condition</i></b> | <b><i>Details</i></b>        |
|------------------------------|------------------------------|
| Internet Speed               | 3G,4G                        |
| Browser Cache                | Disabled                     |
| Device                       | Mobile Simulation on Browser |

|                    |                                                                                                                                  |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------|
| 3d Model File Size | Adidas Stan Smith (9 MB)<br>Cloud-9 Snow (8.3 MB)<br>Nike Air Force One (11.2 MB)<br>Vans Black (5.3 MB)<br>Cloud-9 San (2.4 MB) |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------|

**Table 5 - 15: Average loading time on mobile devices with 3G internet speeds**

| <b>Model</b>                        | <b>Size</b>    | <b>Chrome</b> | <b>Firefox</b>             | <b>Edge</b>   | <b>Safari</b> |
|-------------------------------------|----------------|---------------|----------------------------|---------------|---------------|
| <i>nike-air force-one-plate.glb</i> | <i>11.2 MB</i> | <i>2.7min</i> | <i>2.5min - 1.4 min</i>    | <i>2.8min</i> |               |
| <i>adidas-sam-smith</i>             | <i>9 MB</i>    | <i>2.2min</i> | <i>2.6min<br/>1.3min</i>   | <i>2.2min</i> |               |
| <i>cloud-9-snow-plate.glb</i>       | <i>8.3 MB</i>  | <i>2.7min</i> | <i>2.5min -<br/>1.5min</i> | <i>2.8min</i> |               |
| <i>van-black-plate.glb</i>          | <i>5.3 MB</i>  | <i>1.4min</i> | <i>1.4min - 1 min</i>      | <i>1.4min</i> |               |

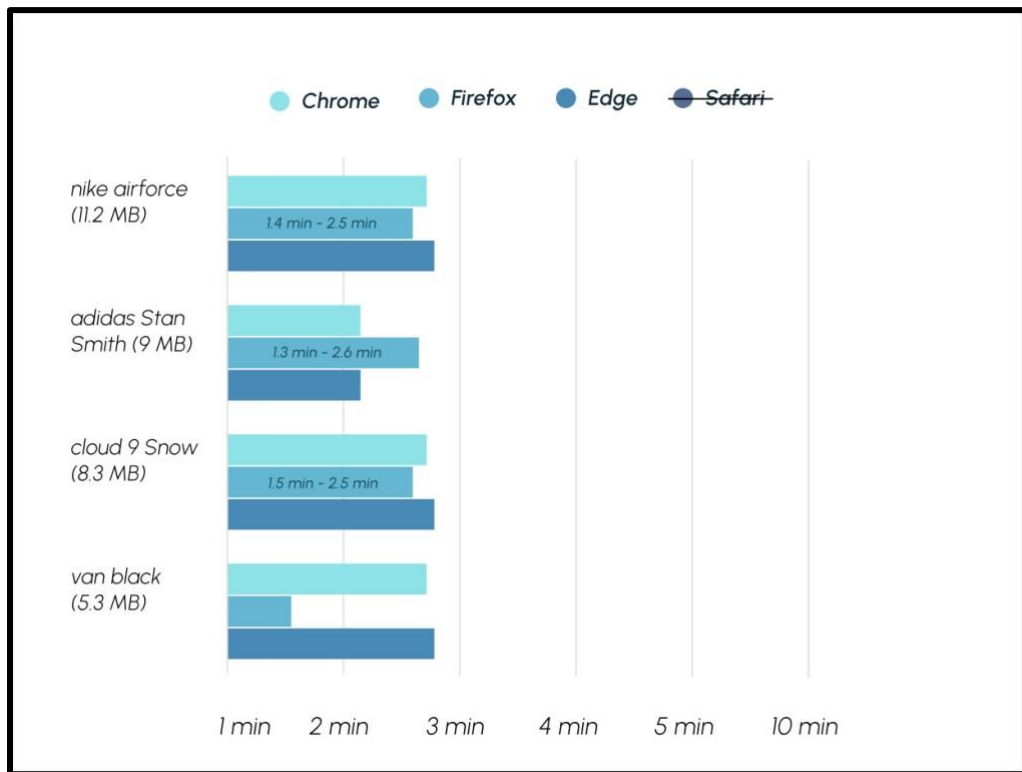


Figure 5.8: Average loading time on mobile devices with 3G internet speeds

Table 5.16: Average loading time on mobile devices with 4G internet speeds

| Model                        | Size    | Chrome | Firefox | Edge | Safari |
|------------------------------|---------|--------|---------|------|--------|
| nike-air force-one-plate.glb | 11.2 MB | 8s     | 16s     | 8s   | 6s     |
| adidas-sam-smith             | 9 MB    | 6.7s   | 14s     | 7s   | 6s     |
| cloud-9-snow-plate.glb       | 8.3 MB  | 8s     | 17s     | 8.2s | 6s     |
| van-black-plate.glb          | 5.3 MB  | 4.38s  | 10s     | 4.4s | 2.6s   |

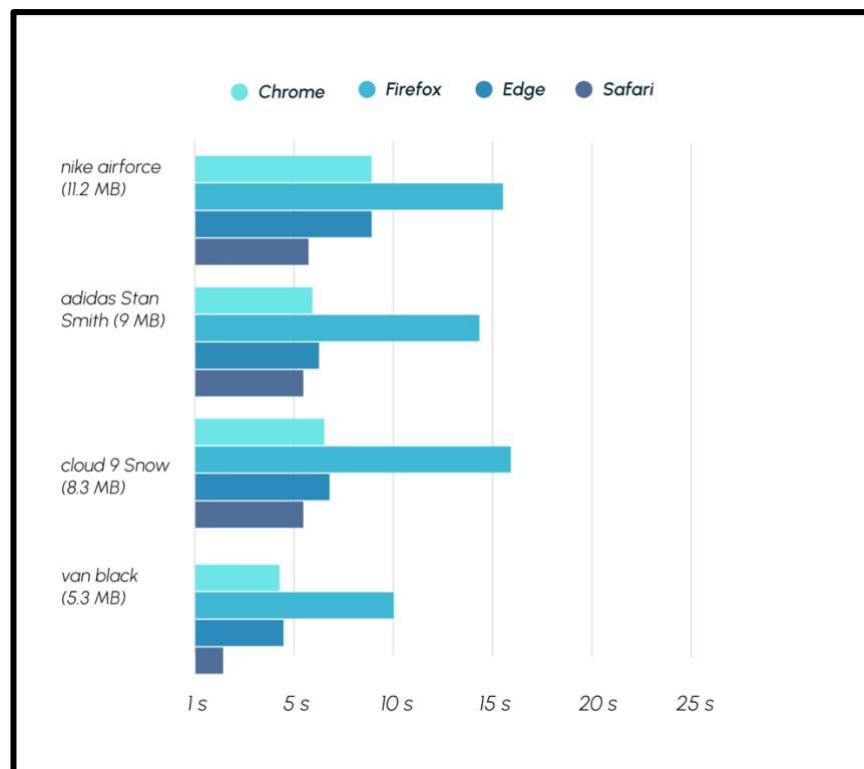


Figure 5.9: Average loading time on mobile devices with 4G internet speeds

**Table 5.17: Total Average loading time on mobile devices with 3G and 4G internet speeds**

| Model                               | Size    | 4G         | 3G              |
|-------------------------------------|---------|------------|-----------------|
| <i>nike-air force-one-plate.glb</i> | 11.2 MB | 6s - 16s   | 4min - 2.7min   |
| <i>adidas-sam-smith</i>             | 9 MB    | 6s - 14s   | 2.8min - 1.4min |
| <i>cloud-9-snow-plate.glb</i>       | 8.3 MB  | 6s - 17s   | 2.8min - 1.5min |
| <i>van-black-plate.glb</i>          | 5.3 MB  | 2.6s - 10s | 1min - 1.4min   |

The findings on the above table suggest that, in addition to reducing file size, developers must optimize delivery methods and test across multiple browsers to ensure consistently acceptable load times under real mobile conditions.

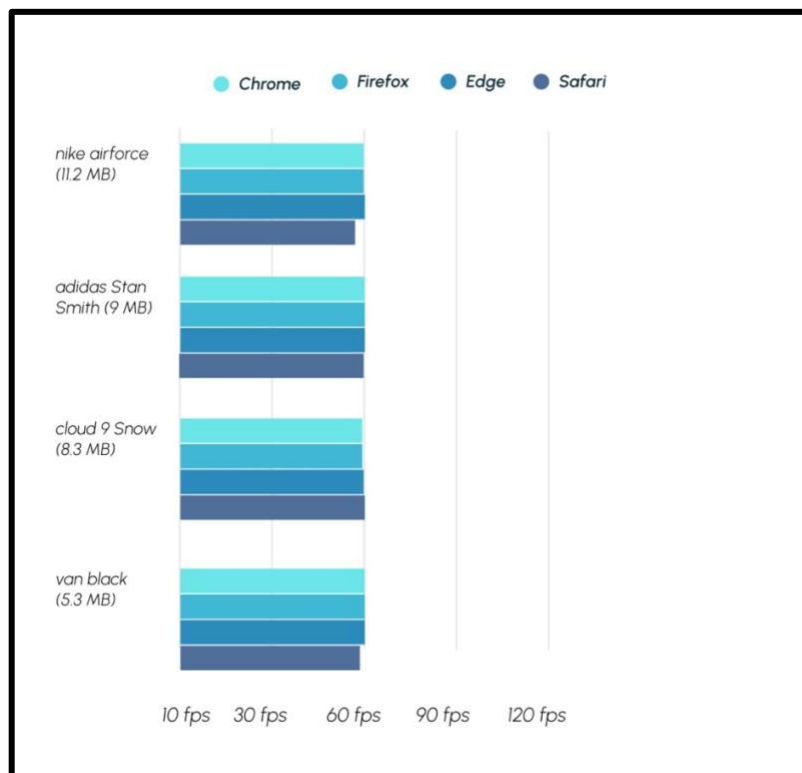
The study found that, larger 3D models took longer to load than smaller ones, but load time did not scale exactly with file size. For example, a mid-sized asset loaded just as slowly as the largest model, indicating that factors beyond raw file weight contribute to delays. Chrome and Edge delivered consistent performance across all models, while Firefox's results varied

more. Safari could not be included in the 3G comparisons because its developer tools lack a built-in 3G network throttling option. These findings suggest that, in addition to reducing model size, developers must optimize delivery methods and test their assets in multiple browsers to ensure reliable load times on slower mobile networks.

### Performance or FPS

**Table 5.18: Summary table comparing the performance metrics from each browser's 3D Model on Mobile Devices**

| Metric                       | Chrome | Firefox | Safari  | Edge   |
|------------------------------|--------|---------|---------|--------|
| nike-air force-one-plate.glb | 60fps  | 60fps   | 56fps   | .60fps |
| adidas-sam-smith             | 60fps  | 60fps   | 60fps   | 60fps  |
| cloud-9-snow-plate.glb       | 60fps  | 60fps   | 60fps   | 60fps  |
| van-black-plate.glb          | 60fps  | 60fps   | 59.7fps | 60fps  |



*Figure 5.10: Average Fps on mobiles' browsers*

### 5.3 Lighthouse Score

Lighthouse is an automated tool that Google uses to score websites on Performance, Accessibility, Best Practices, and SEO. It gives scores between 0 and 100, with larger scores being better. These points will help developers in tracking which areas of improvement needs to be done, particularly with heavy websites containing 3D models whereby visual quality as well as technical performance is important for the best possible user experience across devices and network conditions. The following section presents Lighthouse scores for various 3D model implementations and analyzes their performance characteristics. By examining these metrics, identify optimization opportunities and understand how different 3D models affect overall website performance.

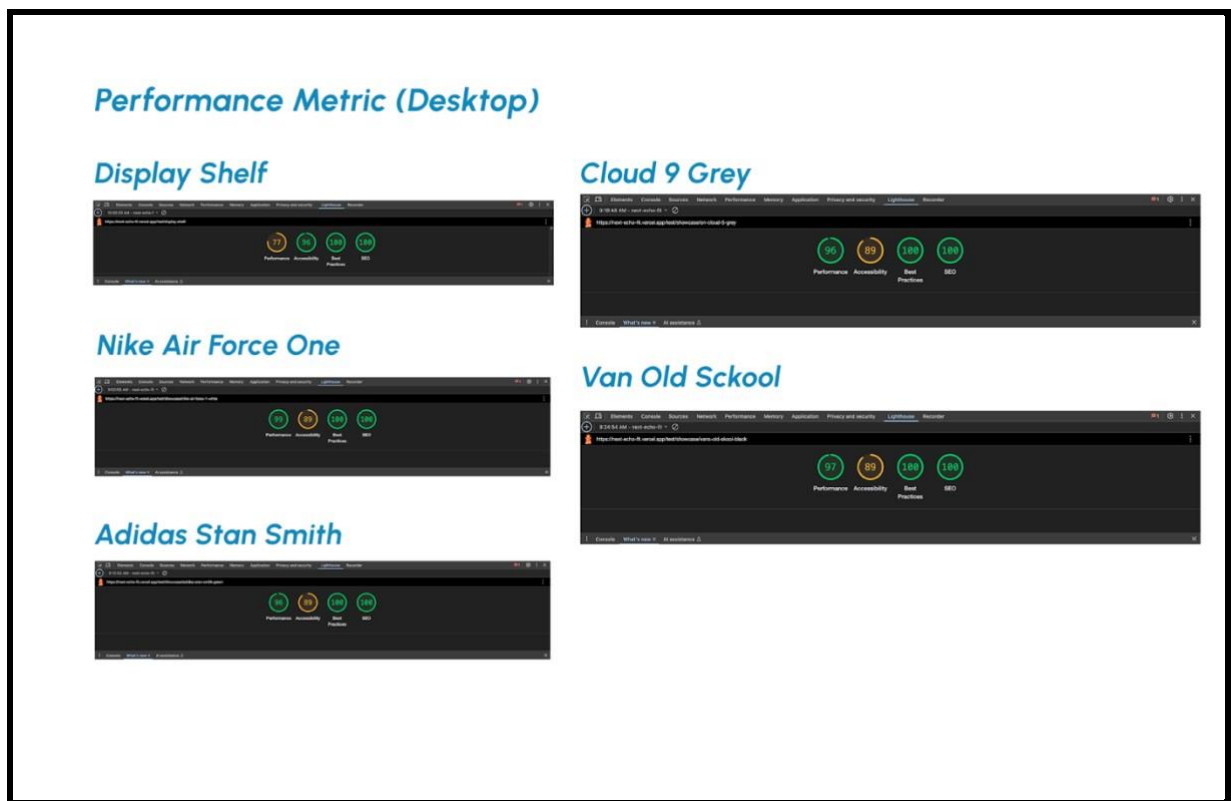
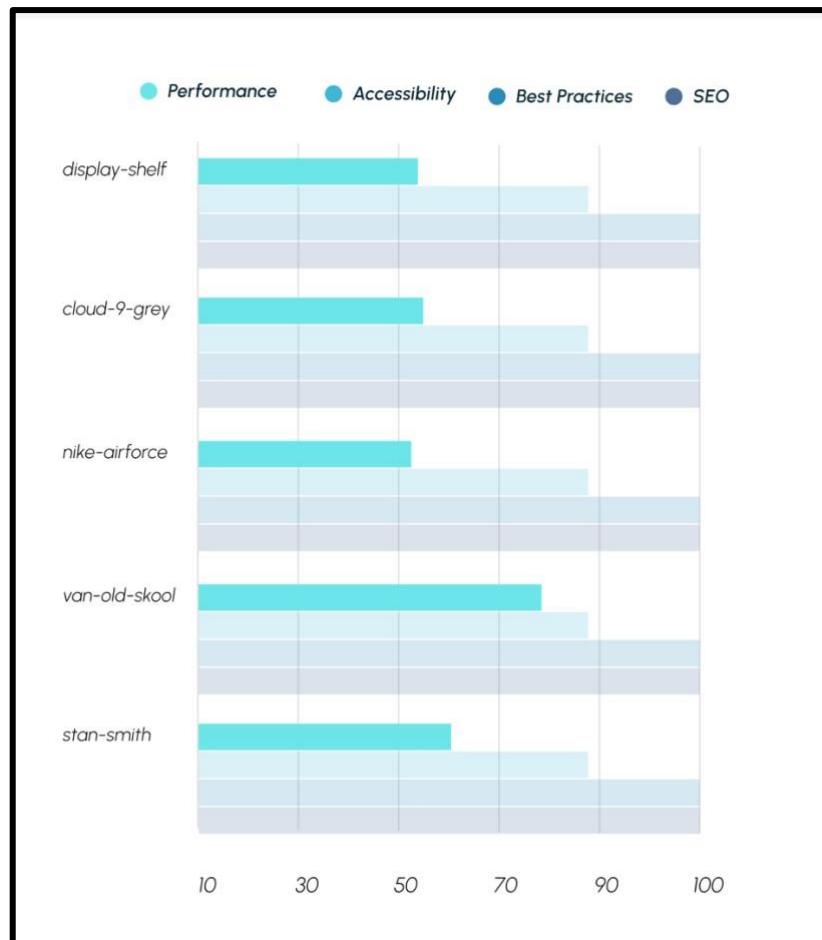


Figure 5.11: Lighthouse Test Results



*Figure 5.12: Lighthouse Test Results in Chart*

The performance metrics for various 3D models provide valuable insights into rendering optimization across desktop environments. While most of the models perform greatly with scores of 96-99, the Display Shelf model falls behind with a score of only 77 on performance. Surprisingly, the Display Shelf model still maintains an exceptionally good runtime performance at a constant rate of 120 fps on desktop browsers discussed in previous sections, suggesting that the low score is mainly due to the high initial loading time instead of rendering inefficiencies.

The contradiction shows that Lighthouse performance scores focus more on how fast a page loads than how well it runs after loading. All models seem to perform consistently very well in Accessibility (89), Best Practices (100), and SEO (100), which shows that they are built on strong fundamentals. Nike Air Force One has the highest performance score of 99, and the optimization techniques learned from it should be applied to the other models. Enhancing Display Shelf would extensively reduce initial load times while keeping its remarkable runtime performance, particularly for mobile adaptation, where hardware constraints are much more significant.

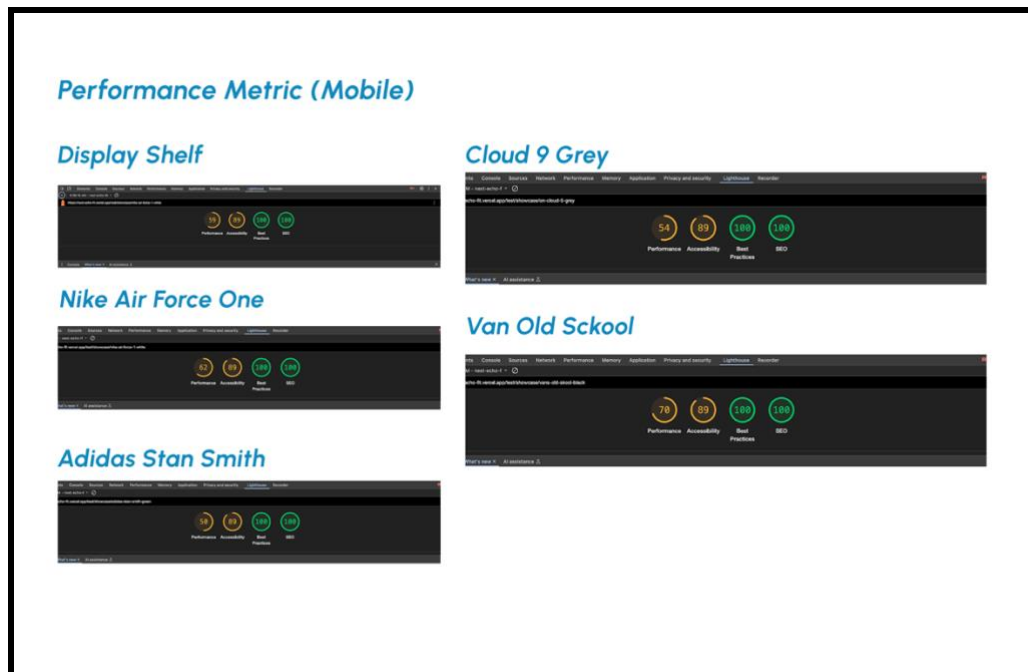


Figure 5.13: Lighthouse Test Results (Mobile)

Table 5.19: Models Lighthouse Score Comparison on Desktop

| 3D Model           | Performance | Accessibility | Best Practices | SEO |
|--------------------|-------------|---------------|----------------|-----|
| Display Shelf      | 53          | 89            | 100            | 100 |
| Cloud 9 Grey       | 54          | 89            | 100            | 100 |
| Nike Air Force One | 52          | 89            | 100            | 100 |
| Van Old Skool      | 78          | 89            | 100            | 100 |
| Adidas Stan Smith  | 58          | 89            | 100            | 100 |

Compared to desktop performance metrics, mobile performance metrics for 3D model viewers show significant optimization challenges. Whereas scores for the desktop fall between 77 and 99, there is a dramatic drop in mobile performance scores, which range between 52 and 78; thus emphasizing the fundamental nature of these challenges which are rendering complex 3D content on less capable devices and possibly slower network connections. The design model Van Old Skool stands firm as the better-performing model for mobile performance at 78, while the other ones, such as Nike Air Force One (52), Cloud 9 Grey (54), Display Shelf (53), and Adidas Stan Smith (58), have all performed poorly in comparison. Nike Air Force One experiences the most extreme shift in performance from best on desktop (99) to worst on mobile (52).

This indicates that the optimization techniques employed are very desktop-specific and do not carry well to mobile environments. Unlike the remaining models, the Nike Air Force One shows strong performance differences, such as its best performance on desktop (99) and terrible in mobile (52). Performance aside, all models maintain a consistency in achieved scores in Accessibility (89), Best Practices (100), and SEO (100), demonstrating that minimum web standards and search optimization practices remain unshaken across device type.

### Final Assessment

| 3d Models         | Loading Time | File Size | FPS | GPU Memory | Lighthouse Score | Memory Usage |
|-------------------|--------------|-----------|-----|------------|------------------|--------------|
| Display Shelves   | ●            | ●         | ●   | ●          | ●                | ●            |
| Cloud 9 Grey      | ●            | ●         | ●   | ●          | ●                | ●            |
| Nike Airforce     | ●            | ●         | ●   | ●          | ●                | ●            |
| Van Old Skool     | ●            | ●         | ●   | ●          | ●                | ●            |
| Adidas Stan Smith | ●            | ●         | ●   | ●          | ●                | ●            |

Figure 5.14: Overall Evaluation Result

## 5.4 Usability Testing

Table 5.20: Summary of Useability Evaluation Process

| Testing Components                      | Description                                                                                                                                                            |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Test Type 1: Task-Based Scenario        | Participants completed similar real-world based e-commerce tasks on their own. (complete breakdown can be seen in table 6.2)                                           |
| Test Type 2: Rating Scale Questionnaire | After test type 1, participants rated their experience using 5-point Likert Scale (1-5) on loading speed, AR smoothness, interface usability, and overall satisfaction |
| Survey Design                           | Nielsen's Usability Heuristics, the System Usability Scale (SUS), and UX best practices were used to develop the questionnaire's usability evaluation criteria.        |

|                                       |                                                                                                                                                                                                                                                    |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Participants                          | A small group of users accessed the AR shoe try-on system independently without assistance to simulate real-world user behavior and reveal usability issues.                                                                                       |
| Recruitment Process                   | Participants were invited to take part in the study through in-person approaches on campus. I introduced the study, explained its purpose, and invited eligible Edinburgh Napier University students to participate directly.                      |
| Research and Informed Consent Process | Explained the research to participants using a participant information sheet, a consent form, and by talking to them in person.                                                                                                                    |
| Selection Criteria                    | Age: 18 years and above.<br>Affiliation: Edinburgh Napier University students.<br>Experience: Familiar with online shopping.<br>Device Access: Smartphone or desktop (for AR testing).<br>Language: Able to understand and communicate in English. |
| Outcome                               | Simple combination of qualitative observations and quantitative data.                                                                                                                                                                              |

**Table 5.21: Test Tasks & Scenarios**

| <b>Task No</b> | <b>Task Scenario</b>                                        | <b>Success Criteria</b>                  | <b>Observation</b>      |
|----------------|-------------------------------------------------------------|------------------------------------------|-------------------------|
| T1             | Visit the homepage and find the virtual shoe try-on feature | User reaches AR try-on page without help | Navigation Clarity      |
| T2             | Activate AR try-on and try to put on your feets             | AR loads correctly without confusion     | Loading time/errors     |
| T3             | Rotate and view the shoe from from different angles         | User successfully interacts with model   | Ease of Interaction     |
| T4             | Add the selected shoe to cart                               | Cart updates correctly                   | Any confusion or error? |
| T5             | Share feedback about the experience                         | User provides honest thoughts            | Satisfaction Level      |

Rating Scale Questions

1. Website Performance & Smoothness \*

Strongly agree

Agree

Neither agree nor disagree

Disagree

Strongly disagree

The website loaded quickly when I opened it.

The AR feature or 3D shoe model loaded without delay.

The AR experience felt smooth without lag or freezing.

Navigating through the website was easy and smooth.

2. Please provide any additional comments or suggestions for Website Performance and Smoothness.

not good

3. AR Try-On and 3D Experience \*

Strongly agree

Agree

Neither agree nor disagree

Disagree

Strongly disagree

The shoe try-on feature was easy to use.

It was easy to rotate, zoom, or interact with the 3D shoe.

The overall design of the website was clear and user-friendly.

4. Please provide any additional comments or suggestions for AR Try-On and 3D Experience

great not bad

5. Overall Experience \*

Strongly agree

Agree

Neither agree nor disagree

Disagree

Strongly disagree

I am satisfied with the overall performance of the website.

I would like to use this AR feature again in the future.

I would recommend this website to others.

Did using the AR feature make the shopping experience more fun or enjoyable for you?

Did using the AR feature help you feel more confident about how the product would look or fit in real life before making a purchase decision

6. Please provide any additional comments or suggestions for Overall Experience

terrible

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Figure 5.15: Rating Scale Questionnaire for AR Shoe Try-On Usability Testing

### 5.4.1 Collected Data Analysis

All of the detailed user responses for Like-Scale Survey can be found in the appendix ([User Like-Scale Rating Questions and Answers](#)).

**Table 5 .22: Task-based Test Result**

| <b>Metric</b>        | <b>Result/Observation</b> |
|----------------------|---------------------------|
| Task Completion Rate | 100%                      |
| Average Time Spent   | 4-6 minutes               |

The results of the usability testing sessions are described in Table 6.2, with an emphasis on important task performance metrics. All participants successfully navigated the website and finished the AR shoe try-on process as intended, as evidenced by the task completion rate of 100%. Activities like browsing, interacting with 3D models, and using the augmented reality feature took up an average of 4 to 6 minutes per user. These findings imply that users were able to effectively interact with the website's features in a reasonable amount of time and that the task flow was generally manageable.

### 5.4.2 Qualitative Insights

In addition to quantitative metrics, qualitative feedback was gathered from participants to provide deeper insights into their experiences, and suggestions while using the AR shoe try-on website. This section captures comments from users and analyzing them, a more comprehensive understanding of user expectations, preferences, and potential areas for improvement is achieved, helping to guide future enhancements of the platform.

#### Website Performance & Smoothness

*"Decent speed", "a bit slow on mobile", "Slight delay on slow 4G", "2-3 mins to load homepage"*

Desktop users gave the load time an average rating of four out of five, which indicates performance that is generally acceptable. However, noticeable delays were reported by mobile users on slower connections (3G/4G). This observation is consistent with technical testing, which revealed that model loading times can surpass two minutes in 3G/4G environments, affecting user perception.

## AR Model Loading & Smoothness

*"Almost instant", "Minor stutter when zooming", "Model needs more realistic style"*

According to user feedback, the AR model loaded nearly immediately and performed well in terms of initial load speed, especially on desktop computers and reliable networks. Although overall performance is satisfactory, a slight stutter was observed during zoom interactions, indicating that interaction-level optimizations are necessary, especially for smoother zoom functionality. This observation is consistent with the findings of technical testing, which showed that during interactive states, some mobile devices and browsers experienced frame rate drops. Users also mentioned that the model needs a more realistic look, suggesting that there is room to improve visual fidelity by adding more realistic textures, lighting, or material details to produce a more realistic and captivating augmented reality experience.

## Navigation Ease

*"A little tricky to find AR section", "Navigation is confusing", "Links should be more standout", "Need to put Back button"*

User feedback revealed navigation challenges within the website. Many participants reported that the navigation was unclear and that links should be more noticeable because they had trouble finding the AR section. One particular suggestion was to make it easier to navigate back to earlier pages or sections by adding a "Back" button. According to these findings, the website's basic functionality works well, but the user experience could be enhanced by making important features more visible, streamlining navigation, and giving interactive elements more obvious visual cues.

## User Interface & Usability

*"Would be great if there's a shoe guide", "Sneakerheads curious with outsole", "Mobile interface needs improvement"*

Some users indicated that they would like to see more additional features, which is adding a shoe guide to offer more through product information. This could be especially helpful for sneakerheads who are interested in particular details like the outsole design. This feedback points

to a chance to add informative or in-depth product content to enhance the AR try-on experience. Furthermore, comments pointed out that the mobile interface needs to be improved, implying that control optimizations or layout changes are required to guarantee a more seamless experience on various devices.

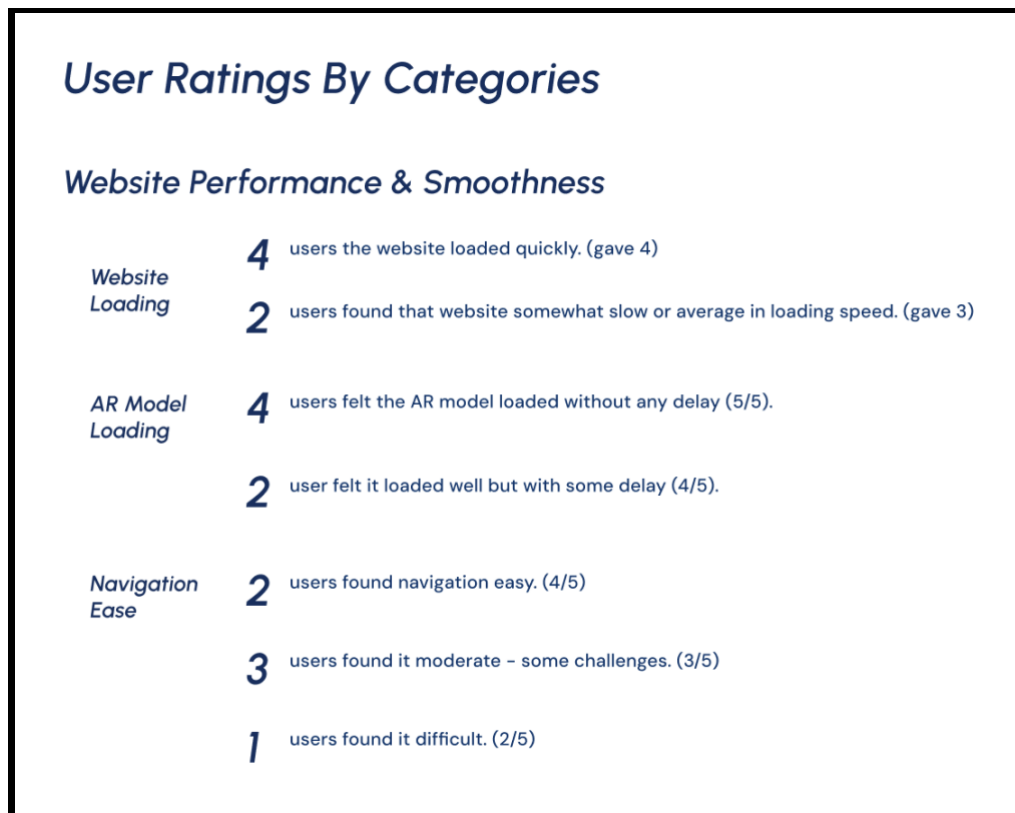
### **Overall Satisfaction & AR Experience**

*"AR needs improvement", "Should be simple and accessible for everyone",*  
*"Need a bit more features"*

Users' comments suggested that although the augmented reality feature worked, its usability and accessibility could be enhanced. Participants stressed that a broader audience, including those who are not as familiar with AR technology, should be able to use the AR experience more easily and intuitively. More features, like better alignment tools, more customization choices, or interactive components, were also suggested as ways to increase engagement. This feedback indicates that the AR interface needs to be improved to maintain an easy-to-use, inclusive, and entertaining experience for all users.

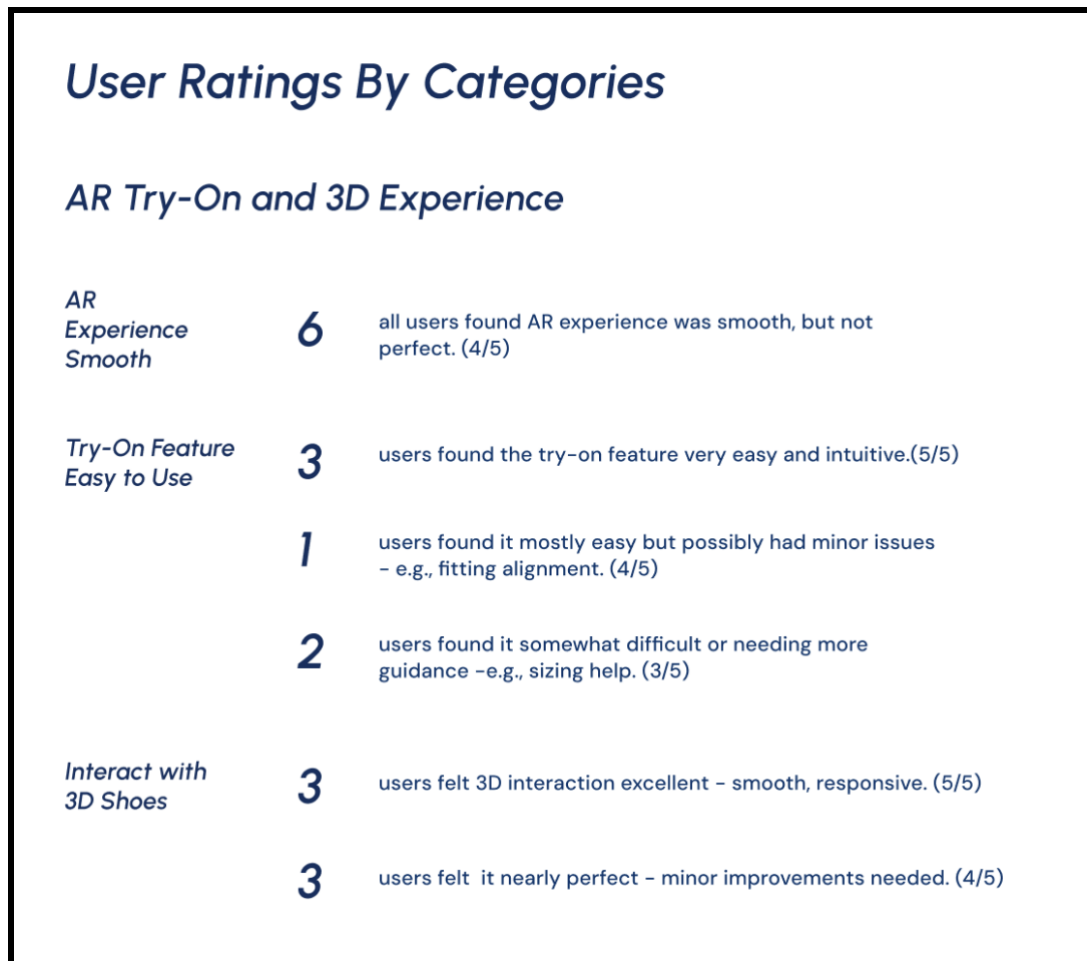
### **5.4.3 Quantitative Data**

This section presents the quantitative analysis of user feedback collected from six participants who evaluated the AR shoe try-on website. The analysis focuses on three core areas: website performance, AR usability, and overall user experience. The data is presented through average ratings and percentage scores, providing insights into user satisfaction and areas requiring improvement.



*Figure 5.16: Users' Ratings by Categories (Performance Section)*

The most positive feedback from users was that the AR try-on and model loading features were easy to use and intuitive. This indicates that the main AR technology is operating effectively. The website's navigation turned out to be the most pain point. Results varied among users, and some found it extremely challenging. It appears that the navigation design is preventing users from enjoying the full AR experience. Performance in the areas indicated by website loading and design clarity was adequate but not exceptional. The technical implementation of AR is good overall, but the website structure that supports it needs to be improved for a smooth user experience.



*Figure 5.17: Users' Ratings by Categories (AR Section)*

All users gave the AR experience overwhelmingly positive reviews, describing it as seamless even though its flaws. This suggests that the fundamental AR features function well in a variety of user scenarios. One notable strength was the 3D shoe interaction, which was rated as excellent by half of the users and nearly perfect by the other half. This implies that the aspects of product visualization are very successful. Results from the try-on feature were more mixed. Half of the users found it to be very intuitive, while others had minor alignment problems or required more help with sizing. This indicates a chance to add better alignment tools or instructions to improve the try-on experience. Though the try-on feature could use some improvement to make sizing and alignment easier, the AR and 3D components function well overall.

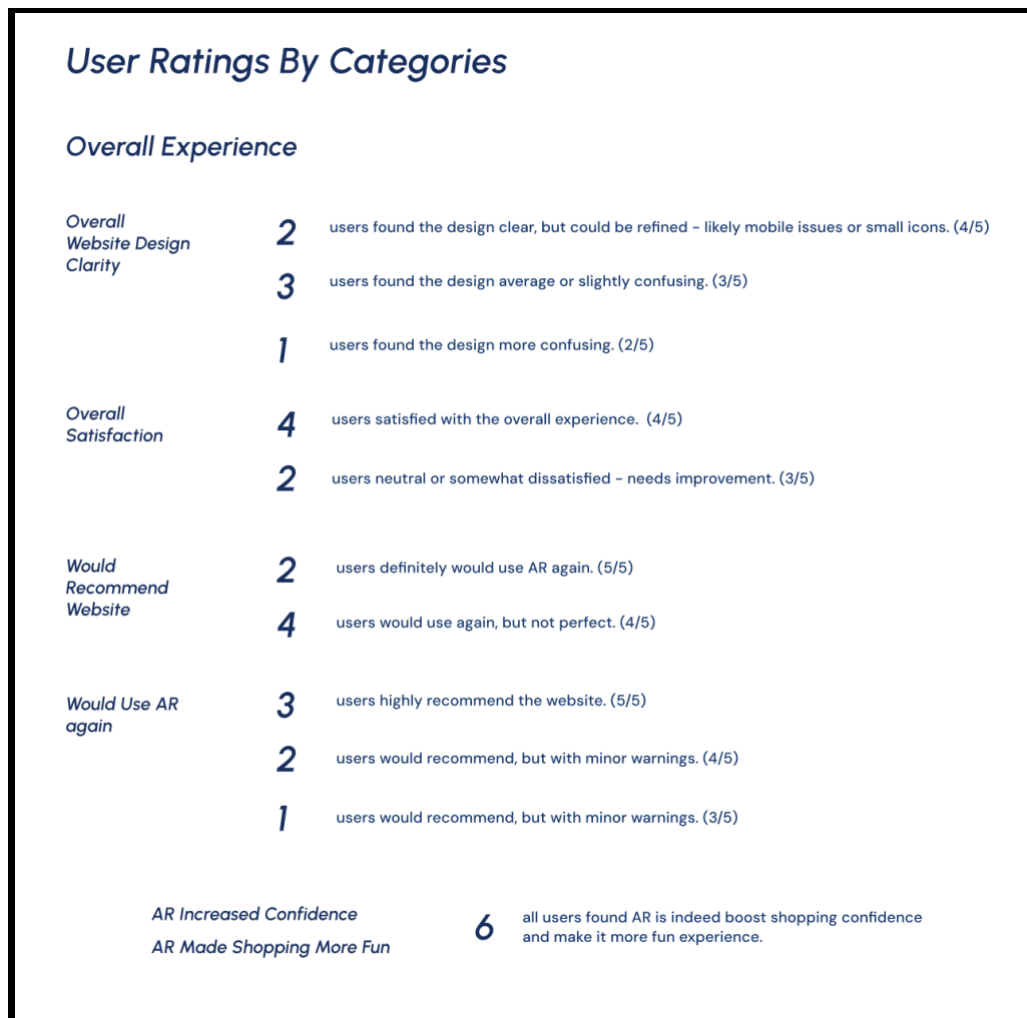


Figure 5.18: Users' Ratings by Categories (Overall Section)

Although website design clarity was identified as a clear area for improvement, the majority of users expressed satisfaction with the overall experience. Most people thought the design was mediocre or a little unclear, with problems that might have been caused by small icons or mobile displays. Every user agreed that augmented reality (AR) technology increased their confidence and enhanced their enjoyment of the shopping experience. This demonstrates how the AR feature achieves its main objective of enhancing purchasing choices. Most users have positive intentions for future use, are willing to use AR again, and recommend the website, with a few slight hesitations. This implies that consumers clearly recognize the value that AR brings to shopping, even while acknowledging some design limitations of the current implementation. While the augmented reality functionality makes for an engaging and empowering shopping experience, improving the website's design clarity would greatly increase user satisfaction and remove any hesitation to return or recommend the service.

To simplify the ratings by users more, all ratings are converted to percentages by using the following formula.

$$\text{Percentage} = \left( \frac{\text{Average Rating}}{\text{Maximum Rating}} \right) \times 100$$

- **Average Rating:** The mean score from your user responses for that category.
- **Maximum Rating:** The highest possible score (in this case, 5).

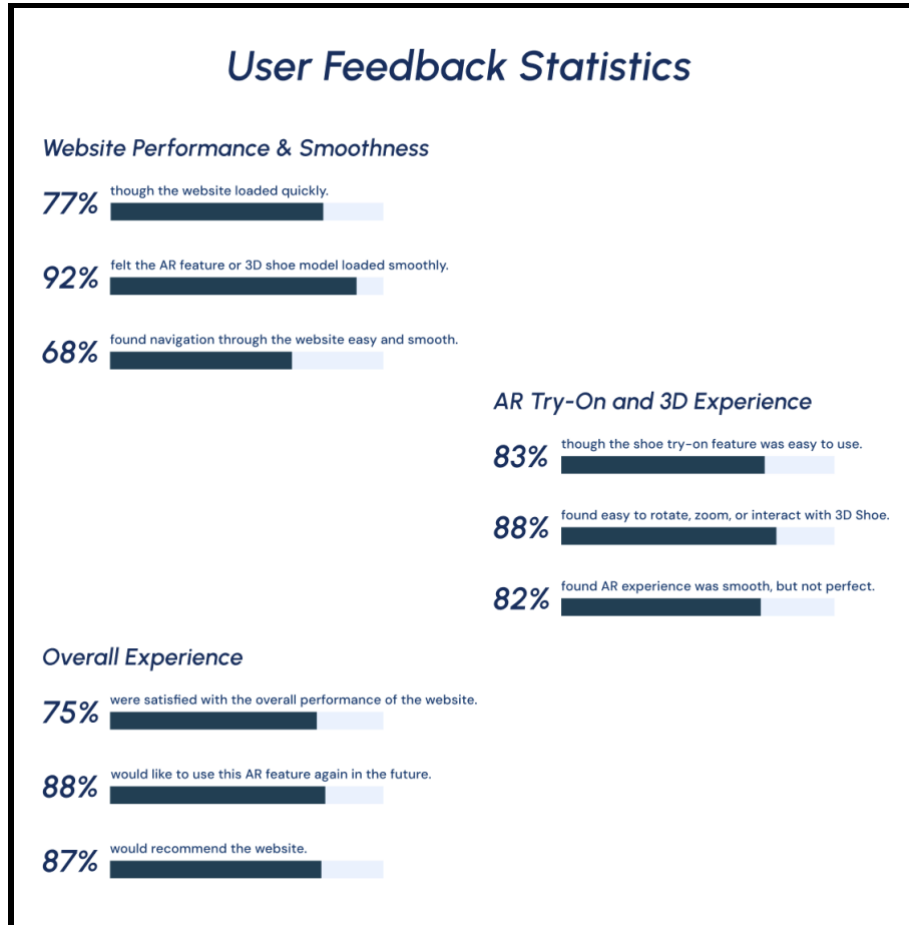


Figure 5.19: Overall User Ratings and Satisfactions

### Average Rating for Website Performance and Smoothness

This component evaluates rating averages by users regarding performance measures, website loading speed, AR model responsiveness, AR experience smoothness, and ease of navigation.

- Website loaded quickly:  $(4 + 3 + 3 + 4 + 4 + 4) = 22 / 6 = 3.67$
- AR model loaded without delay:  $(4 + 4 + 5 + 5 + 5 + 5) = 28 / 6 = 4.67$
- AR experience smooth:  $(4 + 4 + 4 + 4 + 4 + 4) = 24 / 6 = 4.0$
- Navigation ease:  $(3 + 3 + 4 + 3 + 4 + 2) = 19.5 / 6 = 3.17$

$$\text{Average}_{\text{Performance}} = \frac{3.67 + 4.67 + 4.0 + 3.17}{4} = 3.88$$

Figure 5.20: Average Performance Formula

### Average Rating for AR Try-On and 3D Experience

Average user rating for AR Try-On and 3D Experience states how easily users could engage and use try-on, interact with 3D models, and navigate the overall visible design clarity.

- Try-on feature ease:  $(5 + 5 + 3 + 5 + 3 + 4) = 25 / 6 = 4.17$
- Interact with 3D shoe:  $(5 + 4 + 4 + 5 + 5 + 4) = 27 / 6 = 4.5$
- Website design clarity:  $(3 + 4 + 4 + 2 + 3 + 3) = 19 / 6 = 3.17$

$$\text{Average}_{\text{AR Try-On}} = \frac{4.17 + 4.58 + 3.83}{3} = 3.95$$

Figure 5.21: Average AR Try-On Score Formula

### Satisfied with Website Overall (Average)

The average satisfaction reflects an overall impression of website interactions, including performance, usability, and feature engagement.

$$- (4 + 4 + 4 + 4 + 3 + 3) = 22 / 6 = 3.67$$

$$\text{Average}_{\text{Satisfaction}} = 3.67$$

Figure 5.22: Average Satisfaction Score Formula

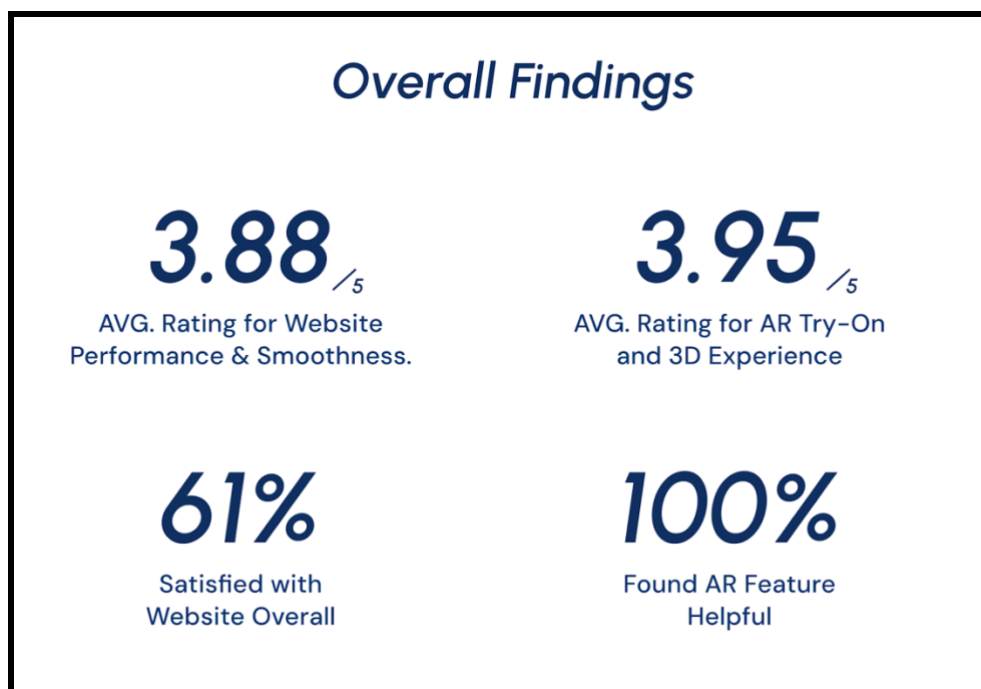


Figure 5.23: Overall User Ratings and Satisfactions

The user study reveals a clear contrast between the AR technology's effectiveness and the website's usability. With a perfect score of 100% of users finding the AR feature helpful for their shopping decisions, the core technology delivers on its promise to enhance the shopping experience. The AR Try-On and 3D features received notably higher ratings (3.95/5) than the website's performance and smoothness (3.88/5).

Navigation and design clarity emerged as the primary areas needing improvement, while AR model loading and 3D shoe interaction were consistently praised. This suggests that while the AR technology itself is strong, the interface through which users access it creates friction in the experience.

With 61% overall satisfaction, there's substantial room for improvement. The data indicates that focusing development efforts on simplifying navigation and clarifying design elements would have the most significant impact on user experience. These improvements would likely eliminate the reluctance some users expressed about recommending or returning to the website.

The study confirms that AR technology for footwear shopping provides genuine value to consumers, boosting their shopping confidence and enjoyment. With targeted refinements to the website framework, this AR application has the potential to deliver a seamless and highly satisfying shopping experience.

## **5.4 Evaluation Conclusion: Key Findings**

After conducting every technical aspect of the project's website, several important findings were identified related to performance, optimization, cost, browser compatibility for web-based 3D and AR experiences.

### **Big File Size**

A major factor identified was the impact of file size and rendering optimization. Large file sizes can significantly delay the download time of 3D models. For example the Display Shelf 3D model which has over 70 MB, indicating unacceptable loading delays, particularly over slower internet connection such as 3g and 4g connections. The evaluation of loading times clearly showed that large file sizes result in noticeable performance issues on mobile devices and lower-speed networks. The bottom line from this to keep file size as low as possible. Even if the file size cannot be reduced, use effective loading techniques including progressive loading.

### **Inconsistent 3D Modeling Approaches**

Another issue is inconsistent modeling techniques, excessive polygon counts, and lack of texture compression. The reasons why inconsistent modeling techniques were discovered is because of the use of 3D models by different artists. CGTrader is an online platform where 3D creators upload and sell their 3D models. Since each artist has their own approach to creative ideas and uniqueness in designing 3D, levels of detail, modeling techniques, texturing, optimization standards, and file structures vary heavily. Furthermore, these models are not designed specifically for websites in mind; they are primarily created for the purpose of showcasing 3d visualizations in the other areas such as product rendering. As a result, it causes inconsistencies in quality and structure, which further led to increased file sizes, inefficient rendering, and increased usage of GPU memory. These issues have a direct effect on performance, especially on mobile devices or slower networks. The bottomline for these challenges is to use either standardized assets or apply the same optimization flow to 3D contents, especially if third-party models are used.

## Cost of High-Quality 3D Model Development

Importantly, another factor to consider is the cost of designing high-quality 3D models. The more detailed and realistic the models are, the higher the costs tend to be. Not to mention, high-detailed models bring the more required resources to create and render, especially when realism and interactivity are desired. The more details, more resources means the larger file sizes will be. The highlights is the need for brands and developers to carefully consider their target platforms, user environments, and directions, before blindly going into development processes. It is essential to maintain balance between visual quality and technical limitations. Not only does this help in managing costs, but also avoiding potential usability and accessibility issues in web-based 3D and AR applications.

## Browser Compatibility and Rendering Performance

The examined results prove that the graphical rendering behavior and loading times vary significantly depending on the browser. Therefore, when optimizing 3D models, developers/designers should keep this in mind during development. Differences in rendering engines and graphics support lead to inconsistent in browser performance. Among tested browsers, Safari was the slowest in terms of responsiveness and compatibility, followed by Chrome, Edge, and Firefox with better results and more reliability overall.

## Mobile Optimization

Moreover, mobile optimization is extremely needed to consider and put more attention into mobile because it is the heart of every AR technology. This evaluation clearly demonstrated lack of mobile optimization in the lighthouse testing section. Although these models scored high on desktop-specified performances, they still failed to satisfy mobile performance benchmarks as evidenced by Lighthouse evaluations. All these described issues align with the factors discussed in the **Literature Review Section 2.8**, including large file sizes, lack of model optimization, and inconsistent asset quality. It is a critical aspect that the focus should be both desktop and mobile devices.

Nonetheless, on the bright side, modern browsers and mobile devices are capable of handling 3D and AR content more smoothly than ever before. With the increasing availability of third-party services and ready-made development tools, implementing WebAR is now more feasible and cost-effective, especially for smaller teams and web projects.

## 6. Conclusions

### 6.1 Main Conclusions

This study explored the development and evaluation of an augmented reality-enabled web (WebAR) platform for virtual footwear try-on, covering the evolution of AR, technical challenges and user experience aspects. The development of WebAR from app- and desktop-based solutions into a browser-based implementation has removed adoption barriers and allowed for wider access and integration into e-commerce platforms. AR indeed improves online shopping in terms of interactivity, emotional engagement, and purchase confidence, which are critical elements in the virtual retail environments. The project has further investigated and shown that introducing AR into e-commerce brings great promise for improving online shopping through immersion, interaction, and reducing uncertainty in consumer purchases and confidence levels.

Some significant issues reported in the technical findings include several effects on the system performance, such as large model file sizes, problems with browser compatibility, and terrible mobile optimization. All these factors make loading times longer, as well as having different experiences over devices, especially on slow networks and weak hardware. However, advanced browsers and mobile devices have become effective and matured to handle and make WebAR possible for retail applications.

User testing and satisfaction surveys confirmed that the principal AR aspects, such as loading, manipulation, and interaction of models, are already well-received and enhance the shoppers' experience. The task completion rate was high (100%), and participants evaluated the system favorably in the usability-related scales. Usability problems concerning interface clarity and navigation, as well as performance lags, especially on mobile devices, were identified as areas that require improvement. Feedback from users indicated that enhancing accessibility and streamlining interaction flows are vital for attracting and retaining users, including those who are less inclined towards AR technology.

Overall, the system has demonstrated the great potential of WebAR in e-commerce. However, for wider acceptance it needs further technical improvement with smaller file sizes, improved browser support, and enhanced responsiveness on mobile devices. In the future, system development should concentrate on further polishing of user interface design, fixing performance, and adding more features for better engagement and satisfaction.

This research contributes valuable insights into the practical implementation of WebAR for retail, reinforcing its capacity to transform online shopping into a more immersive and confidence-inspiring process. Continued innovation and iterative testing are necessary to overcome current limitations and realize the full potential of WebAR in the digital retail landscape.

The research contributes valuable insights into the practical application of WebAR about retail, confirming its potential to enhance online shopping experiences by rendering them more immersive and inspiring confidence. Moreover, there is also a need for continued innovation and

an ongoing testing cycle to address existing limitations and fully leverage the WebAR experience in digital retail.

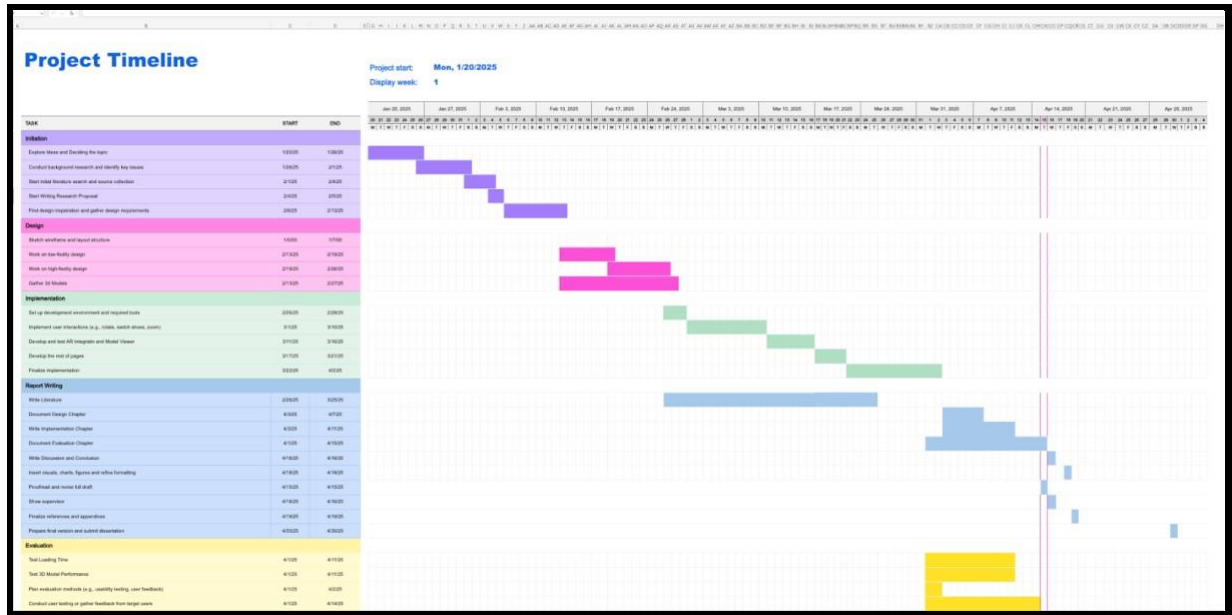
## **6.2 Future Work**

The project discusses a few aspects of augmented reality in relation to online shoe shopping. However, many areas remain unexplored. For example, finding solutions to improve AR work better across a variety of browsers and devices, especially mobile ones and making the loading and compression of 3D models through better optimization and loading methods.

Future research could explore using AI to enable the offering of personalized AR experiences that adapt based on the user, making shopping more engaging and satisfying. There is also a need to consider technical limitations, indicating how foot tracking in virtual try-ons is not perfect yet, and better internet connections for real-time interactions. Close collaboration with the developers and technical experts or 3D designers would be useful in bringing solutions that minimize the technical impediments tied to WebAR.

Finally, testing AR shopping in different retail categories and studying how consumers from various backgrounds respond to it would help us understand how effective WebAR solutions are across different cultures and economic settings.

## Appendix 1 – Project Timeline



| TASK                                                             | START   | END     |
|------------------------------------------------------------------|---------|---------|
| <b>Initiation</b>                                                |         |         |
| Explore Ideas and Deciding the topic                             | 1/20/25 | 1/26/25 |
| Conduct background research and identify key issues              | 1/26/25 | 2/1/25  |
| Start initial literature search and source collection            | 2/1/25  | 2/4/25  |
| Start Writing Research Proposal                                  | 2/4/25  | 2/5/25  |
| Find design inspiration and gather design requirements           | 2/6/25  | 2/13/25 |
| <b>Design</b>                                                    |         |         |
| Sketch wireframe and layout structure                            | 1/0/00  | 1/7/00  |
| Work on low-fidelity design                                      | 2/13/25 | 2/19/25 |
| Work on high-fidelity design                                     | 2/19/25 | 2/26/25 |
| Gather 3d Models                                                 | 2/13/25 | 2/27/25 |
| <b>Implementation</b>                                            |         |         |
| Set up development environment and required tools                | 2/26/25 | 2/28/25 |
| Implement user interactions (e.g., rotate, switch shoes, zoom)   | 3/1/25  | 3/10/25 |
| Develop and test AR Integratin and Model Viewer                  | 3/11/25 | 3/16/25 |
| Develop the rest of pages                                        | 3/17/25 | 3/21/25 |
| Finalize implementation                                          | 3/22/25 | 4/2/25  |
| <b>Report Writing</b>                                            |         |         |
| Write Literature                                                 | 2/26/25 | 3/25/25 |
| Document Design Chapter                                          | 4/3/25  | 4/7/25  |
| Write Implementation Chapter                                     | 4/3/25  | 4/11/25 |
| Document Evaluation Chapter                                      | 4/1/25  | 4/15/25 |
| Write Discussion and Conclusion                                  | 4/16/25 | 4/16/25 |
| Insert visuals, charts, figures and refine formatting            | 4/18/25 | 4/18/25 |
| Proofread and revise full draft                                  | 4/15/25 | 4/15/25 |
| Show supervisor                                                  | 4/16/25 | 4/16/25 |
| Finalize references and appendices                               | 4/19/25 | 4/19/25 |
| Prepare final version and submit dissertation                    | 4/30/25 | 4/30/25 |
| <b>Evaluation</b>                                                |         |         |
| Test Loading Time                                                | 4/1/25  | 4/11/25 |
| Test 3D Model Performance                                        | 4/1/25  | 4/11/25 |
| Plan evaluation methods (e.g., usability testing, user feedback) | 4/1/25  | 4/2/25  |
| Conduct user testing or gather feedback from target users        | 4/1/25  | 4/14/25 |

## Appendix 2 – Project Dairy

EDINBURGH NAPIER UNIVERSITY

SCHOOL OF COMPUTING

PROJECT DIARY

Student: Minn Khant Maung

Supervisor: Ullah, Md Zia

Date: 2/5/2025

Last diary date: First Meeting

Objectives:

- To present the research proposal and get feedback

-

Progress:

- Have gathered some of the articles related to the project.

- Have found to a 3D Designer for shoe display shelf and by the time of writing this, the 3d design is being developed.

- Have obtained resources for implementation of the project

Supervisor's Comments:

- Suggested to set up the overleaf project

- Suggested to create Github Project

- Discussed about the project proposal

- Recommended to start the implementation and finish it as soon as possible

- Suggested to start writing literature review

EDINBURGH NAPIER UNIVERSITY

SCHOOL OF COMPUTING

PROJECT DIARY

**Student:** Minn Khant Maung

**Date:** 2/19/2025

**Supervisor:** Ullah, Md Zia

**Last diary date:** 2/12/2025

**Objectives:**

- Determine next steps for the WebAR implementation phase
- Present partially finished website design and early concept mockups to supervisor
- Review progress on frontend development
- Establish timeline for completing the initial prototype
- Progress of literature review

**Progress:**

- Found some articles for literature review
- Designed 60% of web pages
- Implemented sample website to test 3D was well work with NextJS
- Developed initial frontend for the e-commerce website using Next.js and Tailwind CSS

**Supervisor's Comments:**

- You should begin writing the formal literature review section soon to ensure sufficient time for revisions and refinement
- Consider exploring AI integration with WebAR as this is an emerging area with significant potential.
- Create a GitHub repository for your project code and push regular commits to track progress; likewise, use Overleaf for your dissertation writing to maintain version control
- Make good use of the spreadsheet you've created that summarizes your research article

EDINBURGH NAPIER UNIVERSITY

SCHOOL OF COMPUTING

PROJECT DIARY

Student: Minn Khant Maung

Supervisor: Ullah, Md Zia

Date: 2/26/2025

Last diary date: 2/19/2025

**Objectives:**

- Show completed website design mockups and prototype
- Present progress on frontend development of the e-commerce platform
- Discuss current implementation challenges and ongoing work
- Receive guidance and technical advice from supervisor
- 

**Progress:**

- Completed all website design mockups including product listing, details, cart, and AR interaction screens
- Implemented 60% of frontend functionality using Next.js and Tailwind CSS:
  - Interactive product detail pages with four major control buttons
  - Add-to-cart animation and responsive navigation
  - Integrate designed 3D models to the code and load it to check working or not

**Note:**

- Could not make to the regular meeting due to an emergency circumstance
- Explained the circumstance, reported the progress mentioned above in email and sent it.

**Supervisor's Comments:**

- You are making a good progress.

EDINBURGH NAPIER UNIVERSITY

SCHOOL OF COMPUTING

PROJECT DIARY

Student: Minn Khant Maung

Supervisor: Ullah, Md Zia

Date: 3/5/2025

Last diary date: 2/26/2025

**Objectives:**

- Present progress on frontend development of the e-commerce platform
- Receive guidance and technical advice from supervisor
- Progress on literature review and initial report
- 

**Progress:**

- Implemented 80% of frontend functionality using Next.js and Tailwind CSS:
- Interactive product detail pages with four major control buttons
- Cart system with Zustand state management
- Animation of hover and four modals for four control buttons
- Drafted introduction and technical evolution sections of literature review

 **Supervisor's Comments:**

- Set up Overleaf and GitHub
- Comments on speeding up implementation process
- Include wireframes, UX considerations, user journey in dissertation
- Provide specific completion date for implementation
- Clarify which libraries will be used for WebAR.

EDINBURGH NAPIER UNIVERSITY

SCHOOL OF COMPUTING

PROJECT DIARY

Student: Minn Khant Maung

Supervisor: Ullah, Md Zia

Date: 3/12/2025

Last diary date: 3/5/2025

**Objectives:**

- Present progress on frontend development of the e-commerce platform
- Receive guidance and technical advice from supervisor
- Progress on literature review and dissertation
- Explain current problems and challenges

**Progress:**

- Implemented 86% of frontend functionality using Next.js and Tailwind CSS:
  - Interactive buttons, inputs, and modals in check out page
- Gathered 3d shoe models and formatted to display them on the website
- Discussed WebAR technologies in literature review

**Supervisor's Comments:**

- Overleaf setup and explain how to use
- Show cite in overleaf
- Finish literature review and after that submit initial report

EDINBURGH NAPIER UNIVERSITY

SCHOOL OF COMPUTING

PROJECT DIARY

Student: Minn Khant Maung

Supervisor: Ullah, Md Zia

Date: 3/19/2025

Last diary date: 3/12/2025

**Objectives:**

- To complete remaining website pages and finalize the prototype
- To integrate 3d models into the website using Model Viewer
- To revisit the literature reviews and make some changes

**Progress:**

- Improved UI design for mobile responsiveness.
- Finalized navigation flow and button replacements.
- Conducted final testing on different devices and browsers.

**Supervisor's Comments:**

- Not Attended

EDINBURGH NAPIER UNIVERSITY

SCHOOL OF COMPUTING

PROJECT DIARY

Student: Minn Khant Maung

Supervisor: Ullah, Md Zia

Date: 3/26/2025

Last diary date: 3/19/2025

**Objectives:**

- To complete remaining website pages and finalize the prototype
- To integrate 3d models into the website using Model Viewer
- To complete revisited literature reviews
- To write design chapter, detailing the planning, interface structure, and interaction design of the footwear try-on system, including the design rationale, tools used, and key considerations such as mobile responsiveness and usability.

**Progress:**

- Improved UI design for mobile responsiveness.
- Finalized navigation flow and button replacements.
- Conducted final testing on different devices and browsers.
- Finished writing literature reviews

**Supervisor's Comments:**

- Not Attended

**EDINBURGH NAPIER UNIVERSITY**

**SCHOOL OF COMPUTING**

**PROJECT DIARY**

**Student: Minn Khant Maung**

**Supervisor: Ullah, Md Zia**

**Date: 4/2/2025**

**Last diary date: 3/26/2025**

**Objectives:**

- To write the implementation chapter providing a step-by-step explanation of the development process, system architecture, AR integration, and challenges encountered during the creation of the prototype.
- To complete the Evaluation Chapter, outlining the usability testing strategy, including task completion tests, user engagement metrics, and user satisfaction ratings, and data collected from participants.

**Progress:**

- Documented design process including interface design, user preferences, user interaction, and design considerations.
- Screenshotted designs made in Figma and put them in the design chapter
- Started to work evaluation by setting up necessary tools to evaluate loading times, fps and memory usage.

**Supervisor's Comments:**

- Not Attended

---

EDINBURGH NAPIER UNIVERSITY

SCHOOL OF COMPUTING

PROJECT DIARY

Student: Minn Khant Maung

Supervisor: Ullah, Md Zia

Date: 4/23/2025

Last diary date: 4/16/2025

**Objectives:**

- To insert visuals, charts, figures and formatting
- To finalize references and appendices.
- To prepare final version and submit dissertation

**Progress:**

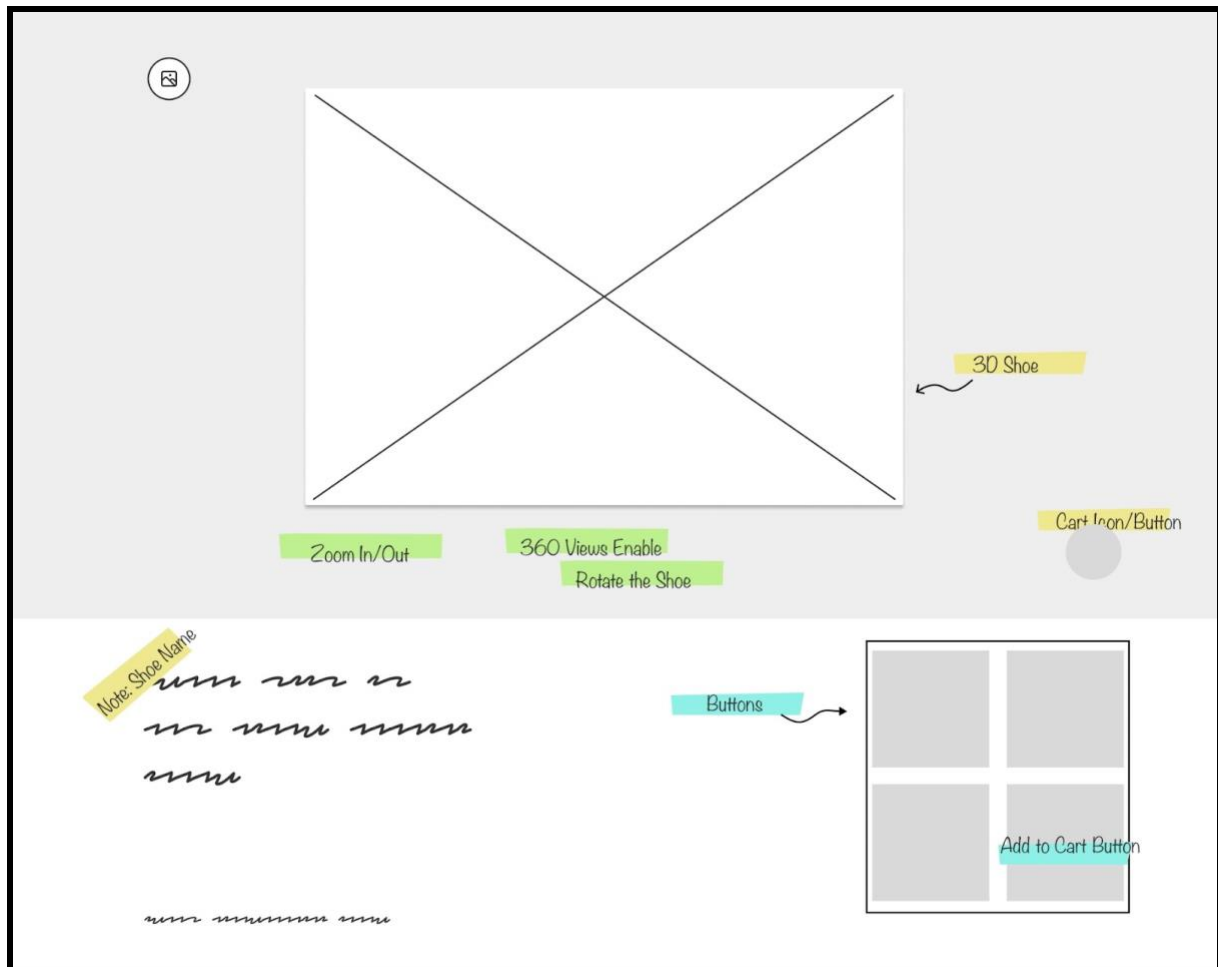
- Documented draft of design, implementation, and evaluation sections.

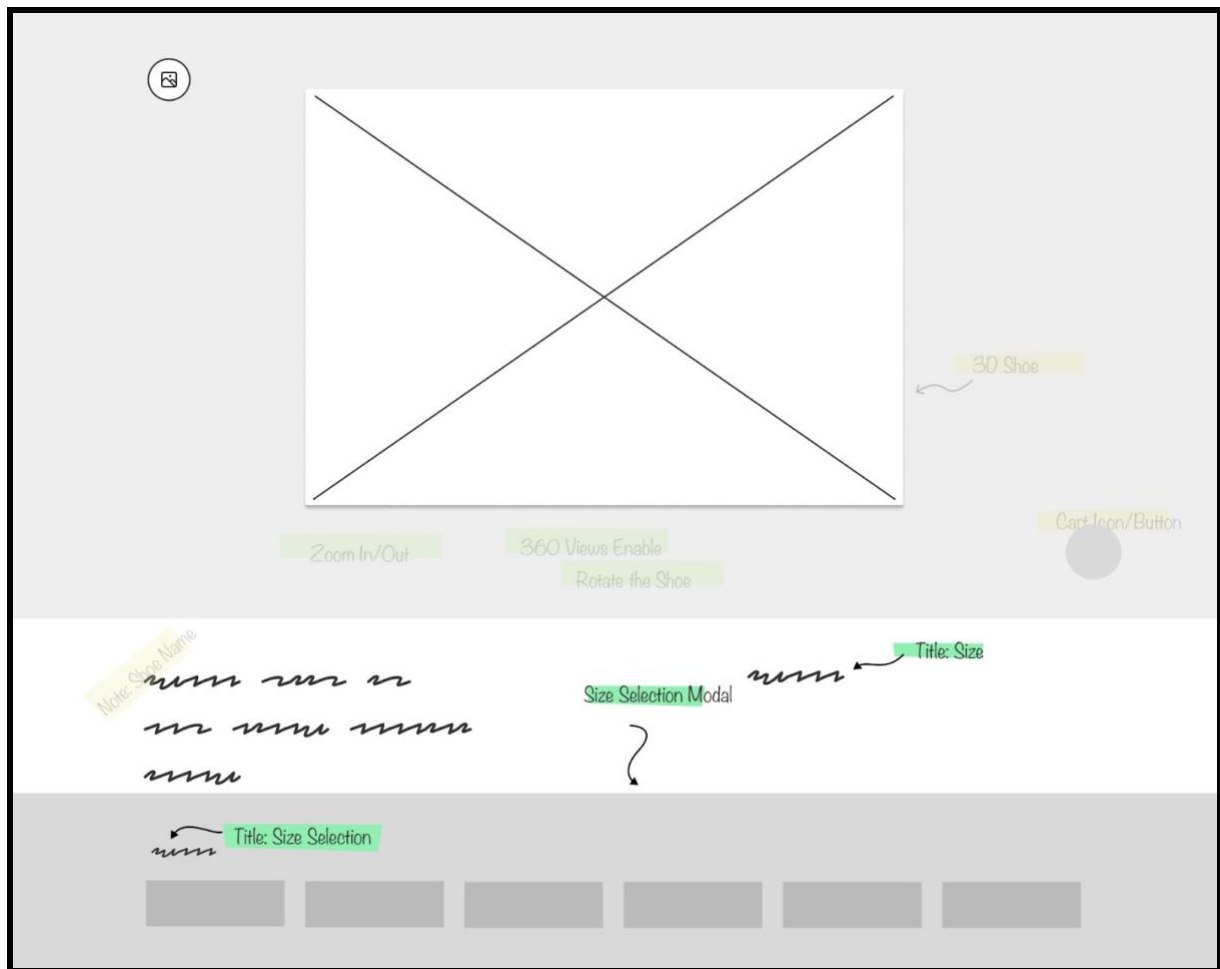
**Supervisor's Comments:**

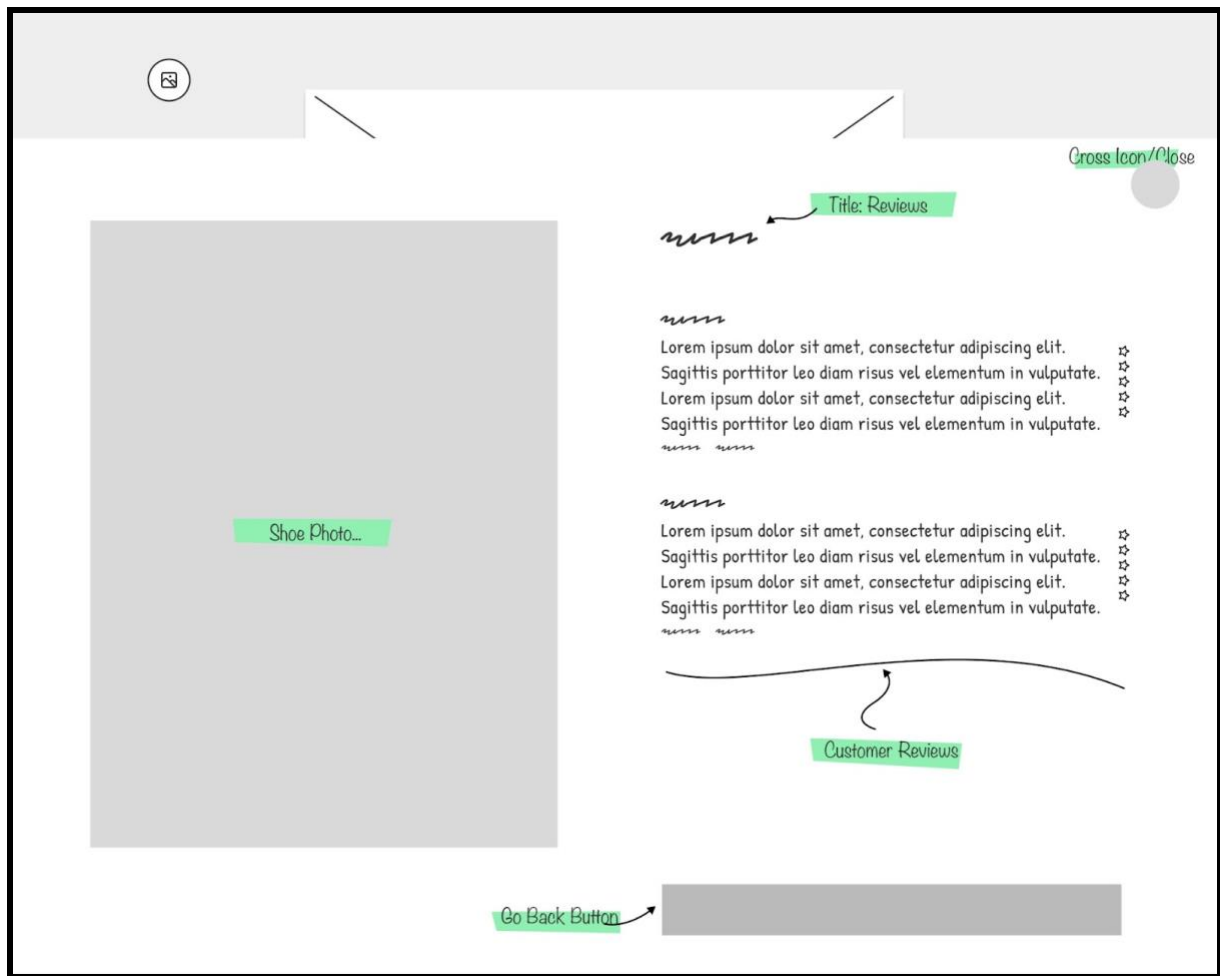


- Not Attended

## Appendix 3 – Low Fidelity Prototype

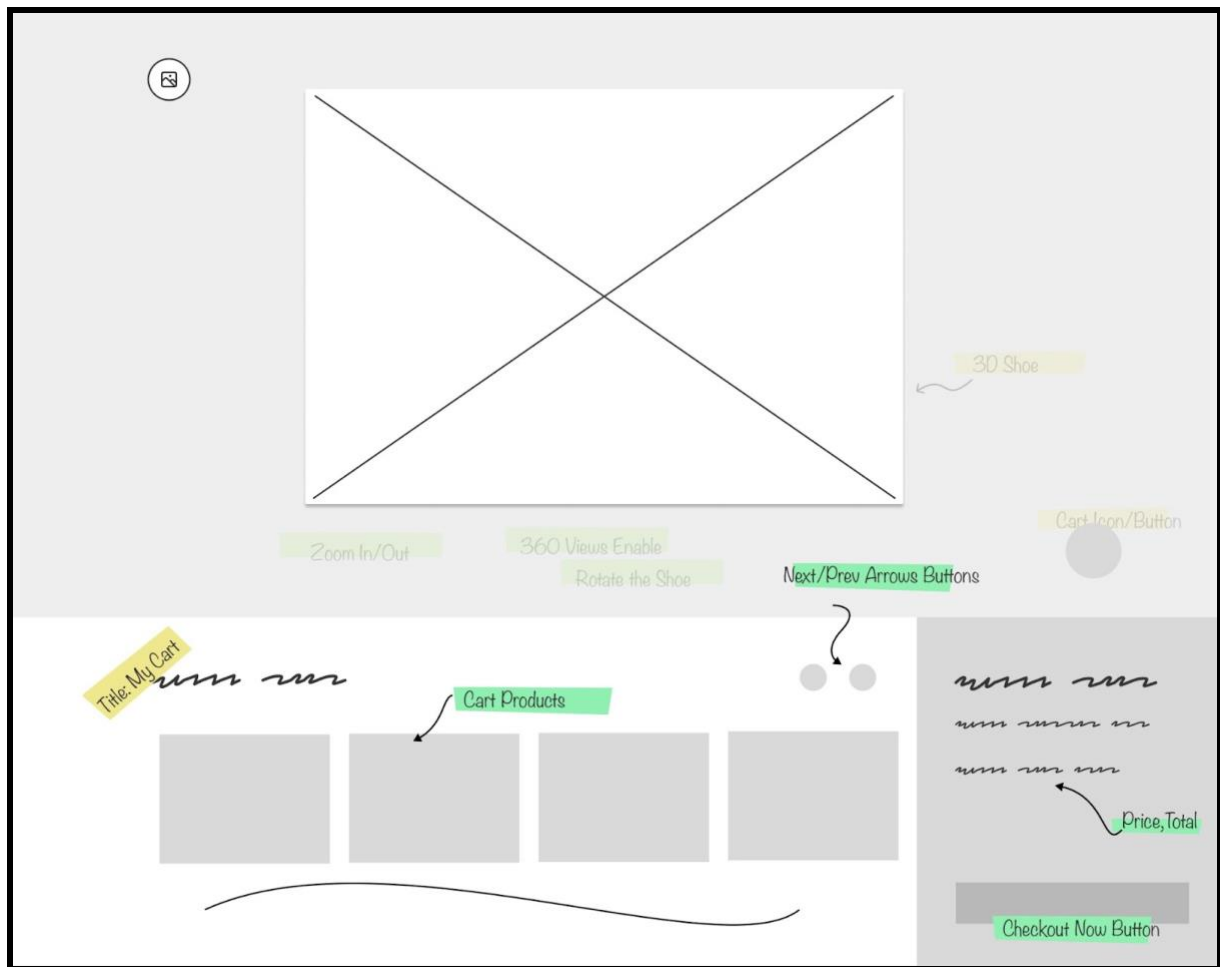






ototype of Size Selection Page







Title: Email



Title: Shipping Address



Title: Delivery Options

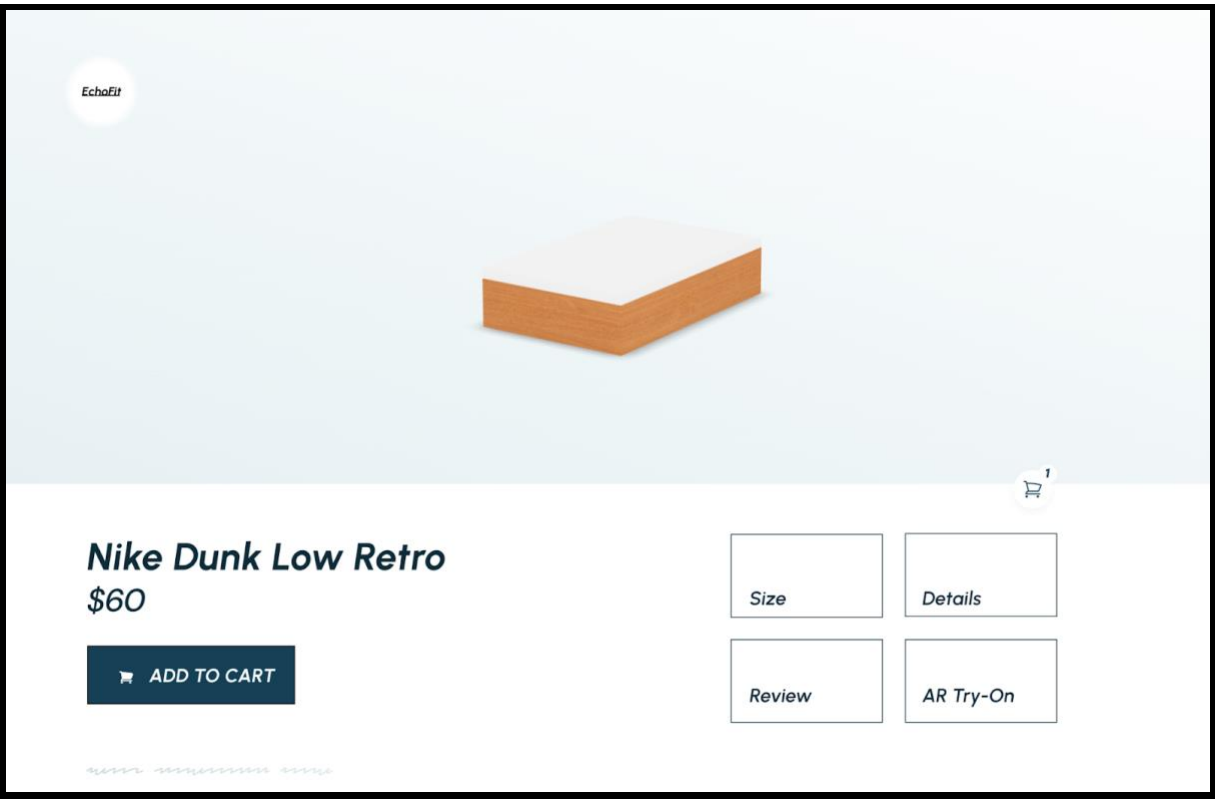
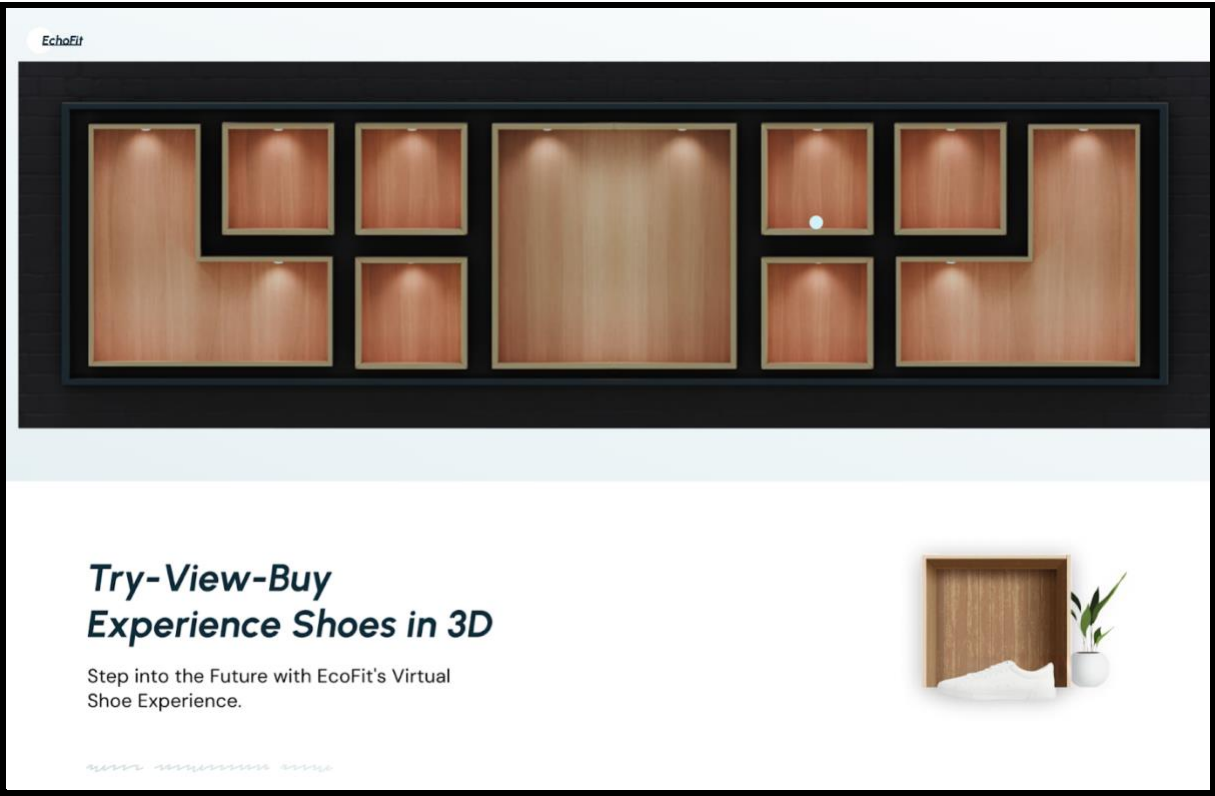


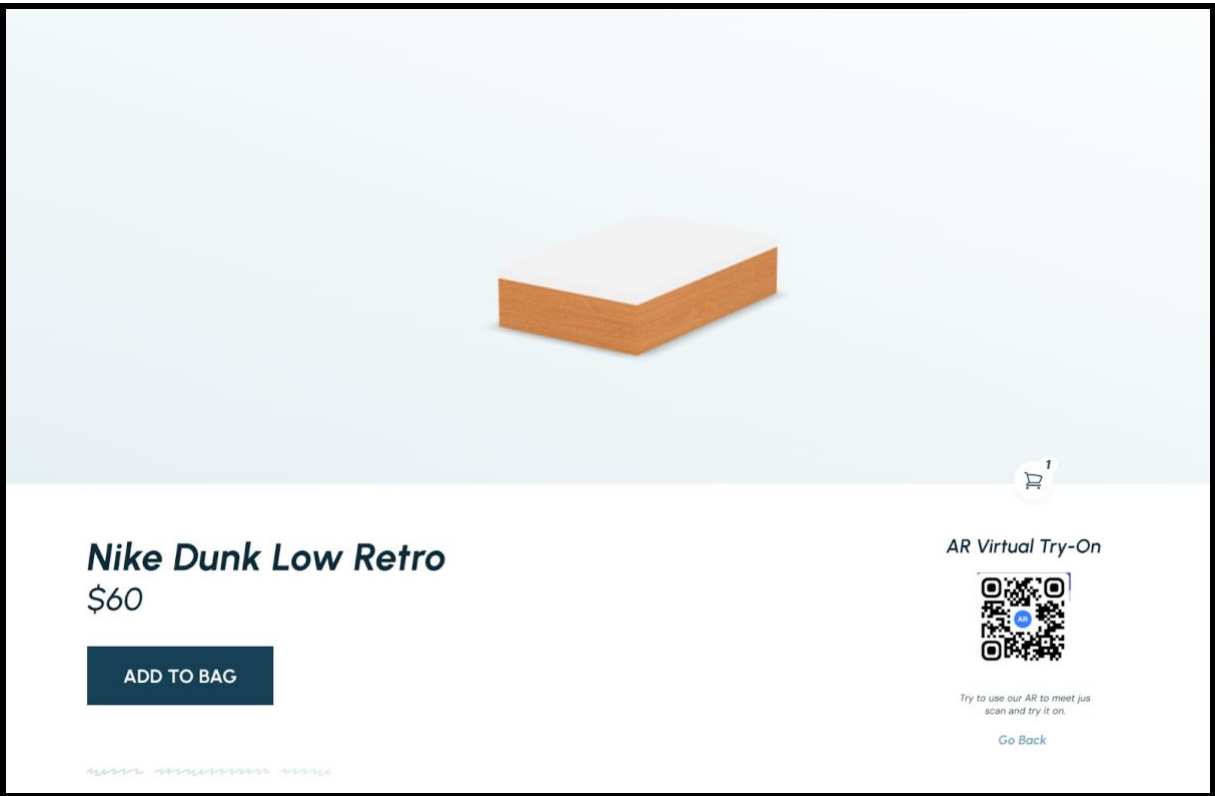
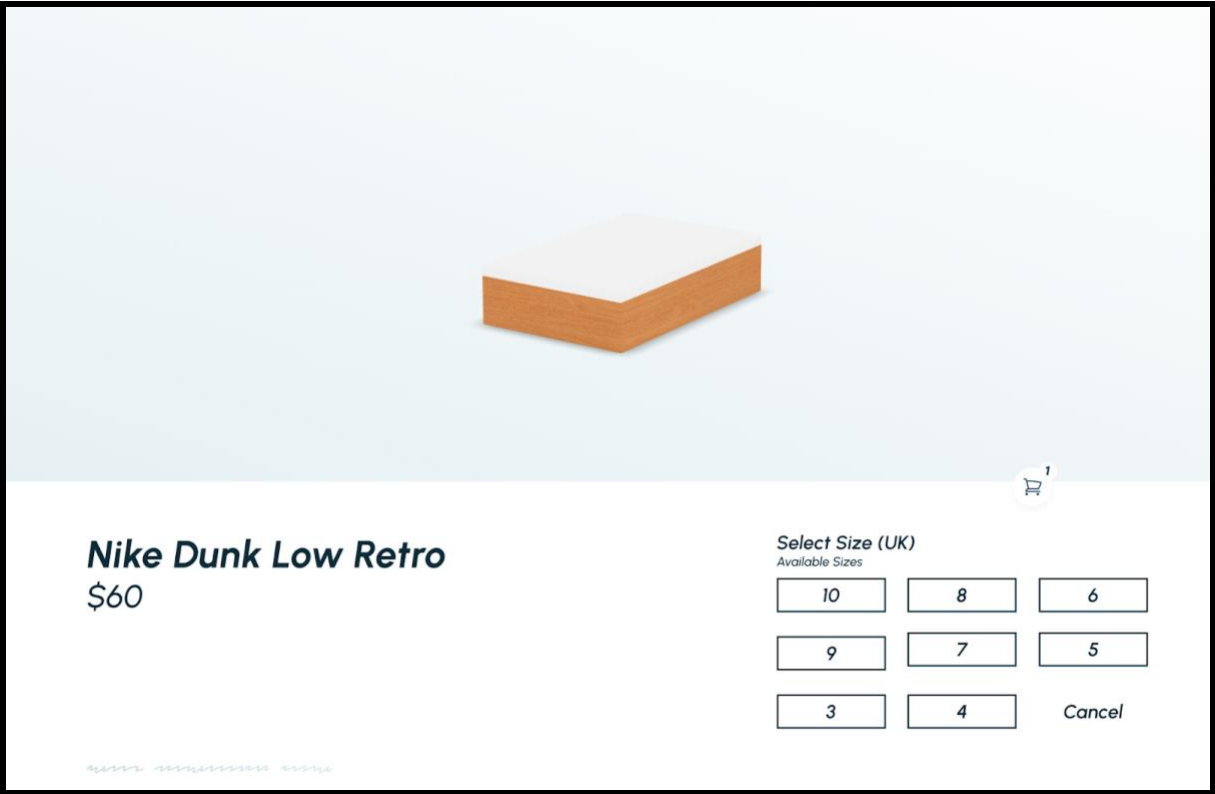
Title: Payment Methods

Pay Now Button

Subtotal, Shipping and Total

# Appendix 4 – High Fidelity Prototype







## Nike Dunk Low Retro

### Best Air Max Ever Made

Recognising the Dunk's roots as the top-ranking university-team sneaker, the Be True To Your School Pack looks to the original ad campaign for inspiration.

RunningMike 15-02-2025



GO BACK

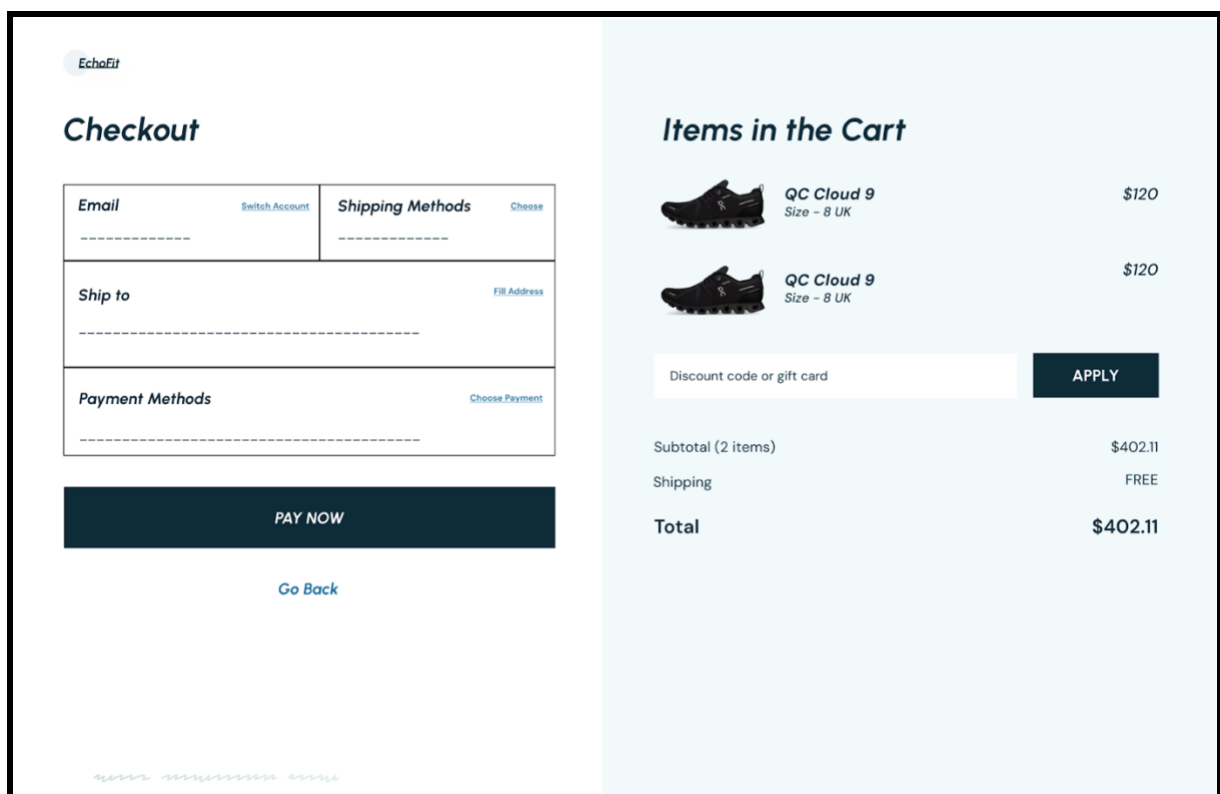
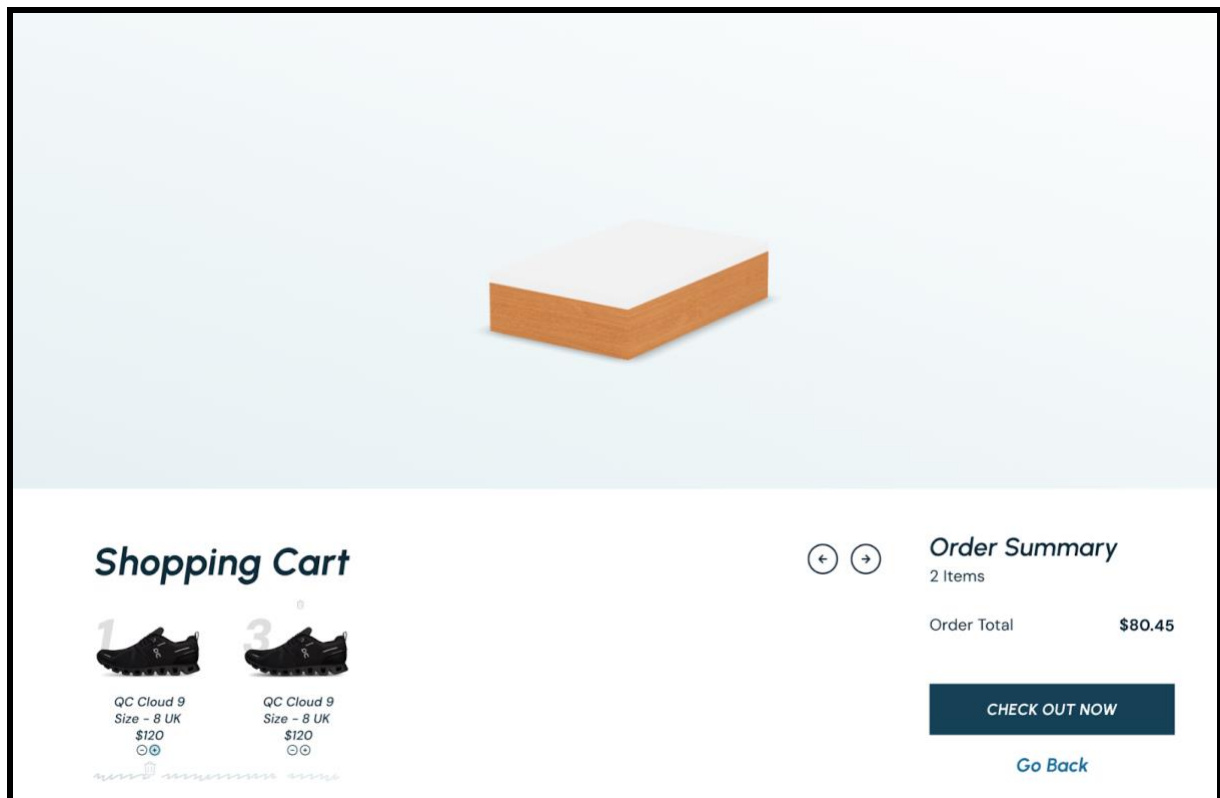


## ***Nike Dunk Low Retro***

Recognising the Dunk's roots as the top-ranking university-team sneaker, the Be True To Your School Pack looks to the original ad campaign for inspiration. Colours represent top-flight universities, while crisp leather has the perfect amount of sheen to make 'em a hands-down win. So lace up and show off that varsity spirit. Are you game?

- Colour Shown: White/Glacier Blue
- Style: DV0833-104

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John

Lastname

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